

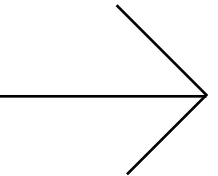


Risk of aquatic invertebrates due to insecticide use in arable crops

xLandscape Case Study

Context: Aquatic Risk Assessment

A classical use of
Landscape Modelling!

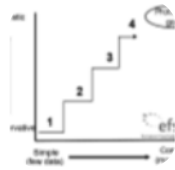


Learning Content permanent VS temporary content

Landscape-level
Aquatic Risk Assessment



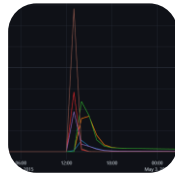
UseCase
Design, Outcome, Communication



Regulatory Framework



xAquaticRisk – Model Approach

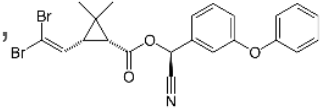


xAquaticRisk – Technical Use



Problem Formulation: lower Tier indicated risk

Pyrethrum is known as insect pest control since antiquity  (Persian insect powder). Synthetic derivatives, **pyrethroids**, appeared in the 1970s, which overcome photoinstability of natural ancestors and which toxicity is selective to arthropods.



There is a high **demand for pyrethroid pest and vector control** (eg, cicada *Philaenus spumarius* spreading *Xylella fastidiosa*, killing olive trees). Pyrethroid pesticides are **authorised in many crops** (eg, cereals, oilseed, soybeans, fruit, olives, cotton, vegetables, rice), in countries across the world.

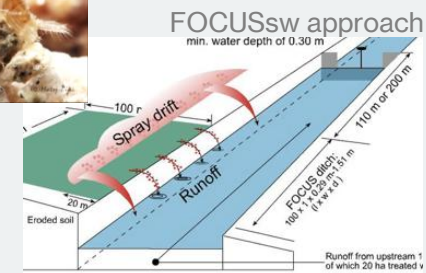
Key environmental concern is the very high toxicity to aquatic invertebrates

We assume a **Regulatory Acceptable Concentration of RAC ~ 1 ng/L**

If exposure, **PEC_{sw} ≤ RAC ⇒ Risk acceptable**

Our CaseStudy '**ModelPyrethroid**' is used in **arable crops**, at an **application rate** of 1x **7.5 g a.s./ha**

Standard, edge-of-the-field exposure assessment results in **PEC_{sw}_FOCUS ~ 2 ng/L → unacceptable risk**



Objectives → More Realistic Risk Assessment for Aquatic Invertebrates

- (1) **Make exposure and effect assessment more realistic**
- (2) **Link effect to exposure modelling.** Identify **effective risk management** measures. Support **weight-of-evidence assessment**.
- (3) **Responsibility and Innovation:** increase knowledge on real-world exposure and effects, also for next generation of approaches.

Supplement

Regulatory Acceptable Concentration (RAC)

The **Regulatory Acceptable Concentration (RAC)** is the **maximum concentration of a pesticide in the environment that is considered safe for non-target organisms**.

At Tier 1 (standard lab tests), the RAC is usually:

Typical steps:

- **Select endpoint: Acute:** LC50/EC50 for fish, invertebrates, algae, macrophytes. **Chronic:** NOEC/EC10 for sensitive taxa.
- **Choose the most sensitive relevant endpoint:** Often the **lowest chronic NOEC/EC10** across taxa is used for long-term RACsw.
- **Assessment factor (AF).** Applied to account for uncertainty: **10** for strong chronic datasets, **50–100** for acute data

$$\text{RAC} = \frac{\text{toxicity endpoint (NOEC, EC10, LC50, etc.)}}{\text{assessment factor}}$$



Higher-tier options (SSD and mesocosms): When Tier-1 is too conservative or fails, EFSA's aquatic guidance allows refinement

- **Species Sensitivity Distributions (SSD):** Use multiple chronic endpoints (e.g. EC10/NOEC for many species). Fit an SSD and derive an **HC5** (hazard concentration for 5% of species). Apply a smaller AF (e.g. 1–5) to HC5 → refined RAC.
- **Micro-/mesocosm studies (model ecosystems):** Expose realistic aquatic communities to the pesticide. Identify the **highest concentration with negligible or acceptable population-level effects**, depending on the protection goal. This concentration (possibly with a small AF) becomes the **RAC**
- **ETO vs ERO (protection goal choice).** EFSA distinguishes two protection goal options for deriving RACs from higher-tier data:
 - **Ecological Threshold Option (ETO):** Only **negligible effects** at population level are allowed. RAC is set at the highest concentration with no or only minor, transient effects.
 - **Ecological Recovery Option (ERO):** Some initial population-level effects are allowed **if recovery occurs within an acceptable time** (e.g. within weeks). RAC can be higher than the ETO-RAC, but must be justified by clear recovery in the mesocosm data.

Once the aquatic RAC_{surfacewater/sediment} is set, it is compared with modelled or measured exposure: eg, **PEC_{sw} ≤ RAC_{sw} → aquatic risk acceptable**

If the risk is not acceptable, regulators may require risk mitigation (eg, buffer zones, drift-reducing nozzles, reduced application rates) or they may reject the product.

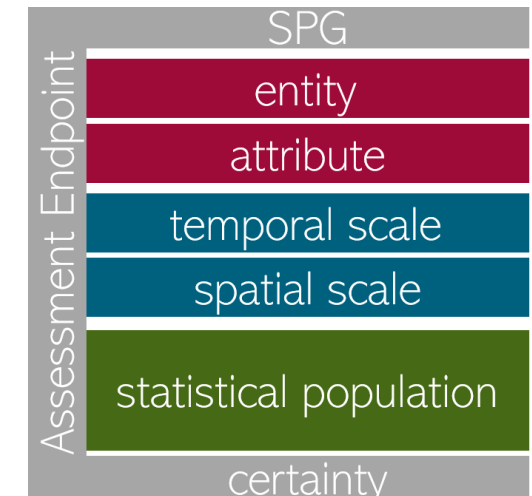
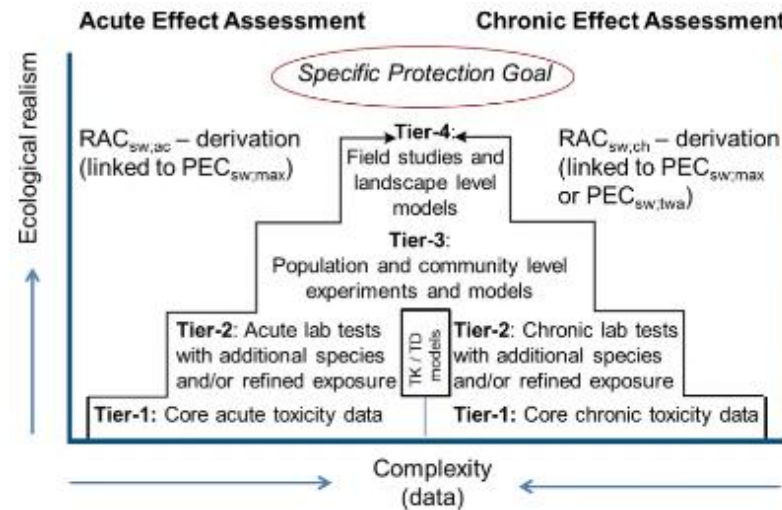
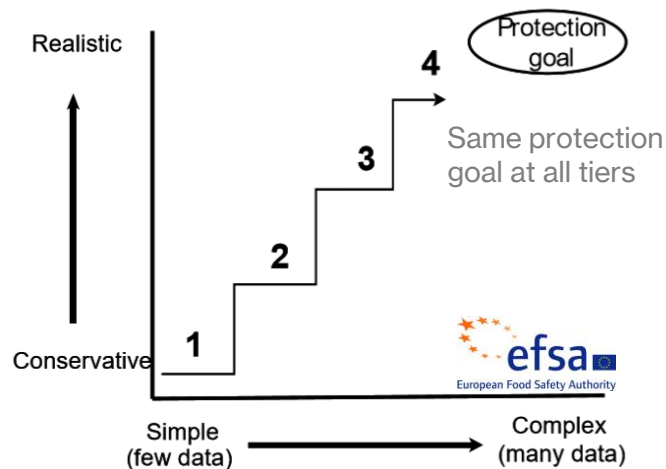
At lower (standard) tiers, PEC_{sw/sed} are calculated using the FOCUS_{sw} approach, at higher-tier landscape models and scenarios come in, eg, xAquaticRisk

why can we do that?

Tiered Aquatic Risk Assessment (EU)

Objective: Higher-Tier Approach guided by Specific Protection Goals

Assessment: acceptable effects? → Quantify Exposure and Effects in **Dimensions** and **Scales** defined in **SPGs**



*SPG=Specific Protection Goal

Regulation EC 1107/2009, Data Requirements, Uniform Principles, **Aquatic Guidance Document**, Sustainable Use Directive, EFSA Scientific Opinions on **Specific Protection Goals**, PPR Panel scientific opinions on **recovery** in aquatic organisms/ecological **threshold** concepts, EFSA/PPR **opinions and guidance on TKTD**, EFSA **uncertainty + weight-of-evidence guidance**, **FOCUS Surface Water**, ...

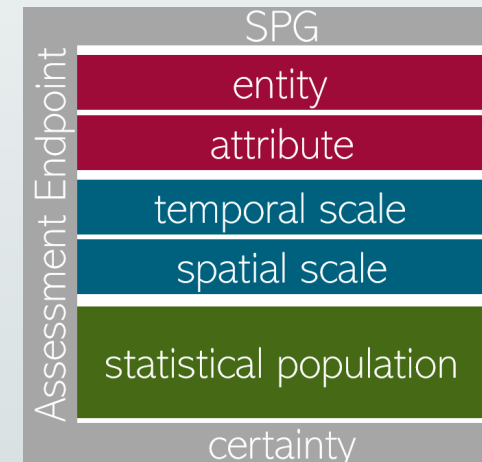
Supplement

Specific Protection Goals

Basis: Specific Protection Goals (SPGs) for aquatic invertebrates, Aquatic Guidance, TKTD GD, ...

EFSA defines (SPGs) for aquatic invertebrates by linking ecological functions (below) to population-level endpoints:

- **Entity** to protect: **Populations** of aquatic invertebrates (not individuals)
- **Attribute** to protect: **Abundance** and structure (e.g. species composition)
- **Magnitude** of effect: **either negligible** (Ecological Threshold Option, ETO) **or recoverable** (Ecological Recovery Option, ERO)
- **Temporal scale**: Recovery within a defined time frame (e.g. 8 weeks)
- **Spatial scale**: Edge-of-field surface waters (e.g. ditches, ponds, streams)



SPGs are based on key roles of aquatic invertebrates in surface water:

- Decomposition and nutrient cycling
- Food web support (e.g. prey for fish and amphibians)
- Water quality regulation

"Protecting populations ensures these functions remain intact even if some individuals are affected" (EFSA)

EFSA classifies aquatic invertebrates as one of the **key drivers** of ecosystem services in aquatic systems. SPGs for this group are therefore set at the **population level or higher**, because maintaining their ecological functions (e.g., nutrient cycling, food-web support) requires populations to remain intact.

Supplement

Risk vs Vulnerability

Concept	What it fundamentally represents	Core question it answers	Typical data sources	Regulatory role
Risk	Probability and magnitude of adverse effects given exposure	“How likely is harm to occur under defined use conditions?”	Exposure models, toxicity endpoints, residue data, application scenarios	Central metric for authorization decisions
Vulnerability	Intrinsic and extrinsic susceptibility of a species or population to harm	“How sensitive or exposed is this species/population in its ecological context?”	Life-history traits, habitat use, phenology, landscape structure, stressors	Increasingly used for refinement, mitigation, and landscape-level protection goals

let's build a Tier-4

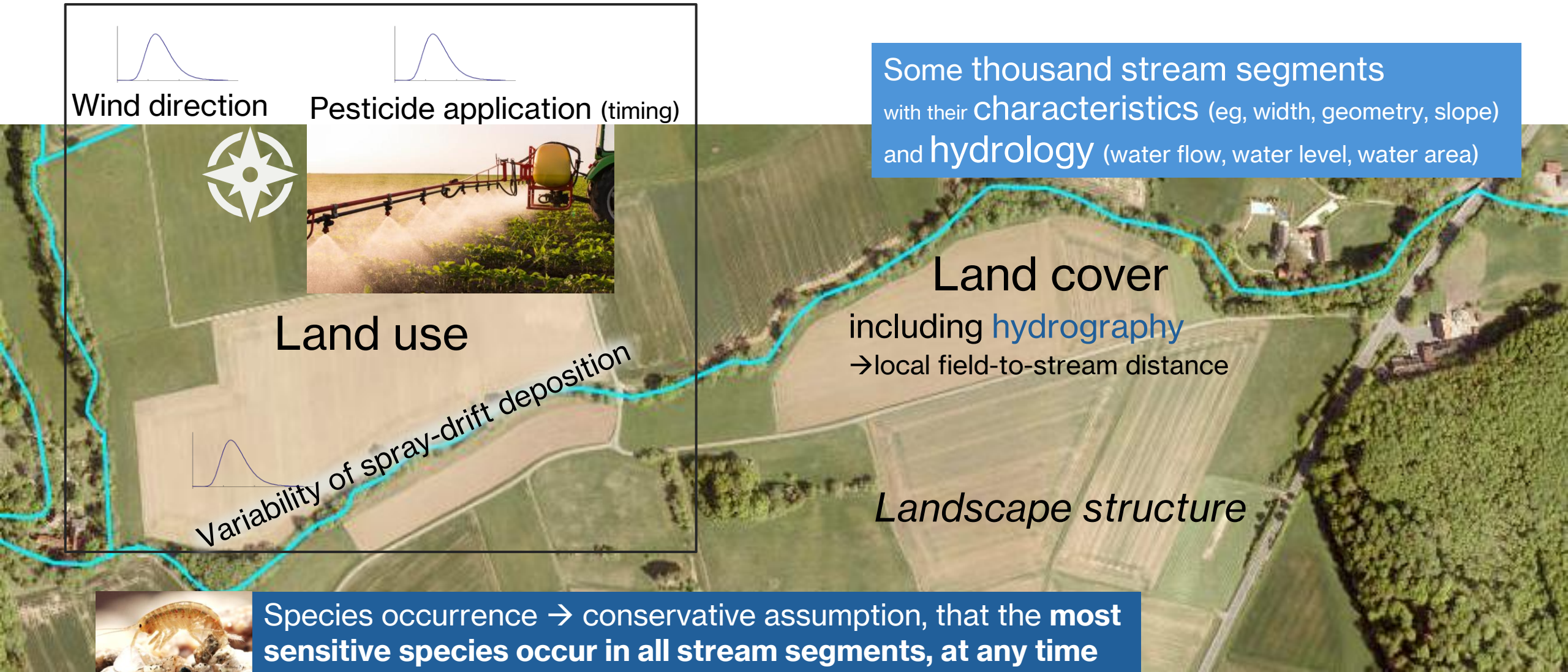
how to design 'more realistic'



- 1. Which factors drive exposure**
of aquatic systems by the ModelPyrethroid?
(common approaches: expert judgement, literature, sensitivity analysis)
- 2. Which factors drive risk**
of aquatic invertebrates by the ModelPyrethroid?
(aspects: species ecotoxicological sensitivity, ecological sensitivity, life cycle, occurrence, habitat type and quality, eg, water body type and conditions, drying out)
- 3. Which factors to refine** (in a first iteration),
considering data availability, robustness, preparation effort, accessibility, cost, comprehensibility, transparency, coverage, scope/transferability, ...
(eg, environmental factors, land use, pesticide application, risk management)
- 4. Which risk mitigation** (risk management) to consider (in a first iteration)
(eg, buffer distance, drift-reducing technology, ...)
5. How would you design an **improved Tier-4** landscape level risk assessment approach for aquatic invertebrates? (answer to: thorsten.schad@landwerk-ev.de)

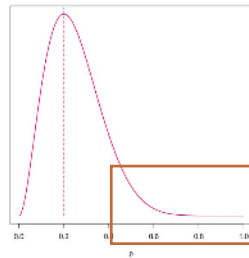
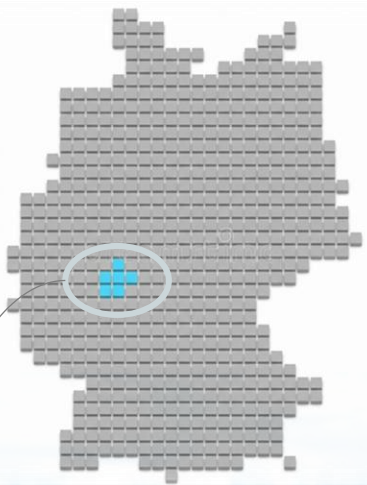


Landscape factors we make more realistic



Real-world System Variability

Let's assume that the **authorisation unit** is a **country**, eg, Germany



Variability of risk accross the authorisation unit

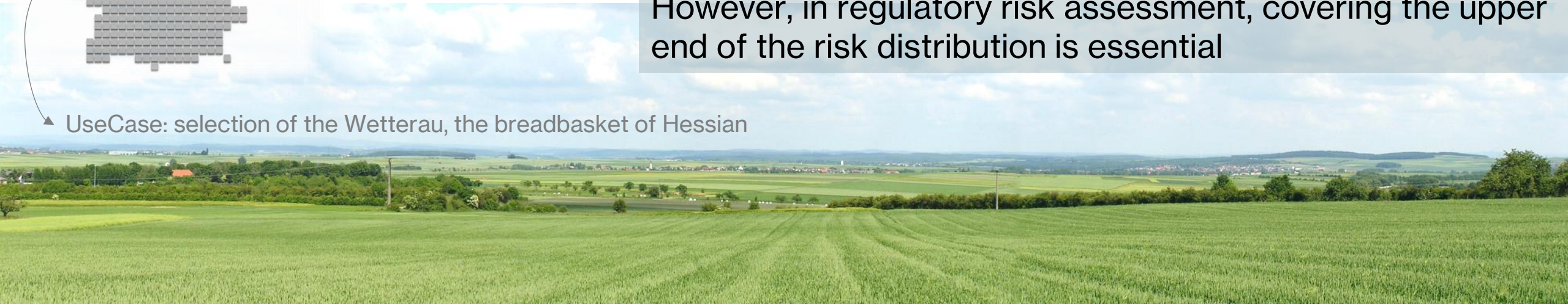
A landscape approach has to **address the range of risk occurrences of the decision making domain**. This, **spatially** (geographically):

- 1) **Full coverage** (currently possible for regional crops, eg, hops, orchards, vegetables)
- 2) **Sampling**, starting with the selection of study sites (scenarios), as a step to discretising variability

and **temporally**, by using long-term records

However, in regulatory risk assessment, covering the upper end of the risk distribution is essential

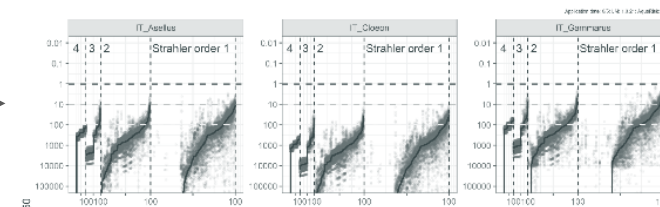
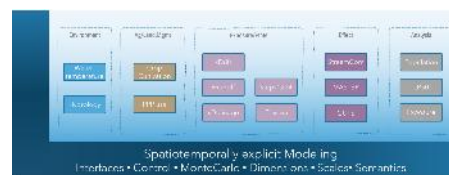
▲ UseCase: selection of the Wetterau, the breadbasket of Hessian





Real-world System Variability

Propagate real-world variability to model prediction



Hybrid Model (xLandscape)

- Discretisation (eg, space, time)
- Monte Carlo (PDFs*)

Input variability, eg

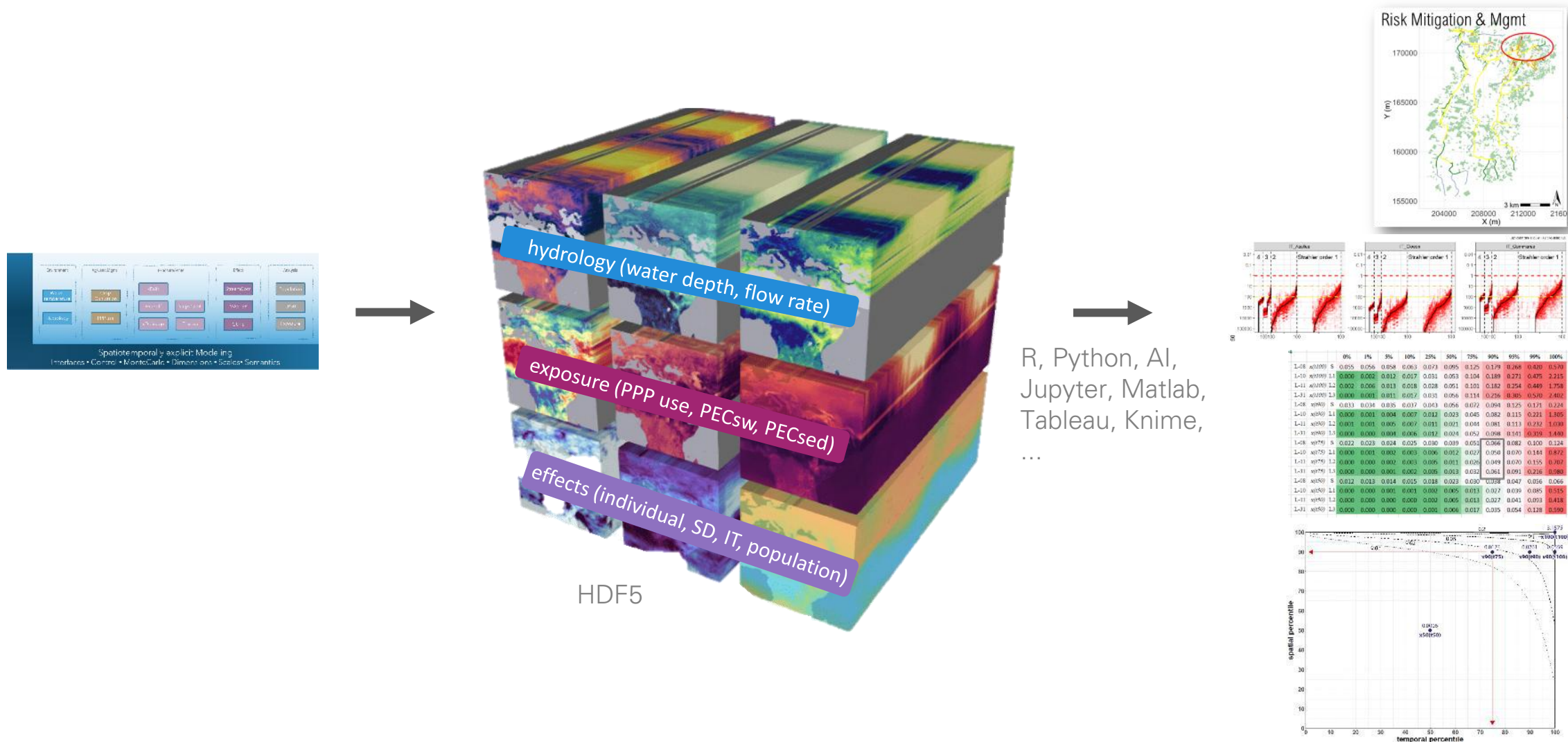
- › Agriculture: land use, pesticide use
- › Environment: weather, soil, habitat conditions
- › Biology: species occurrence, life cycle

Output variability, eg

- › Exposure: PECs, residues
- › Effects: mortality, growth, population size
- › Environment: bee forage

*PDF: Probability Density Function

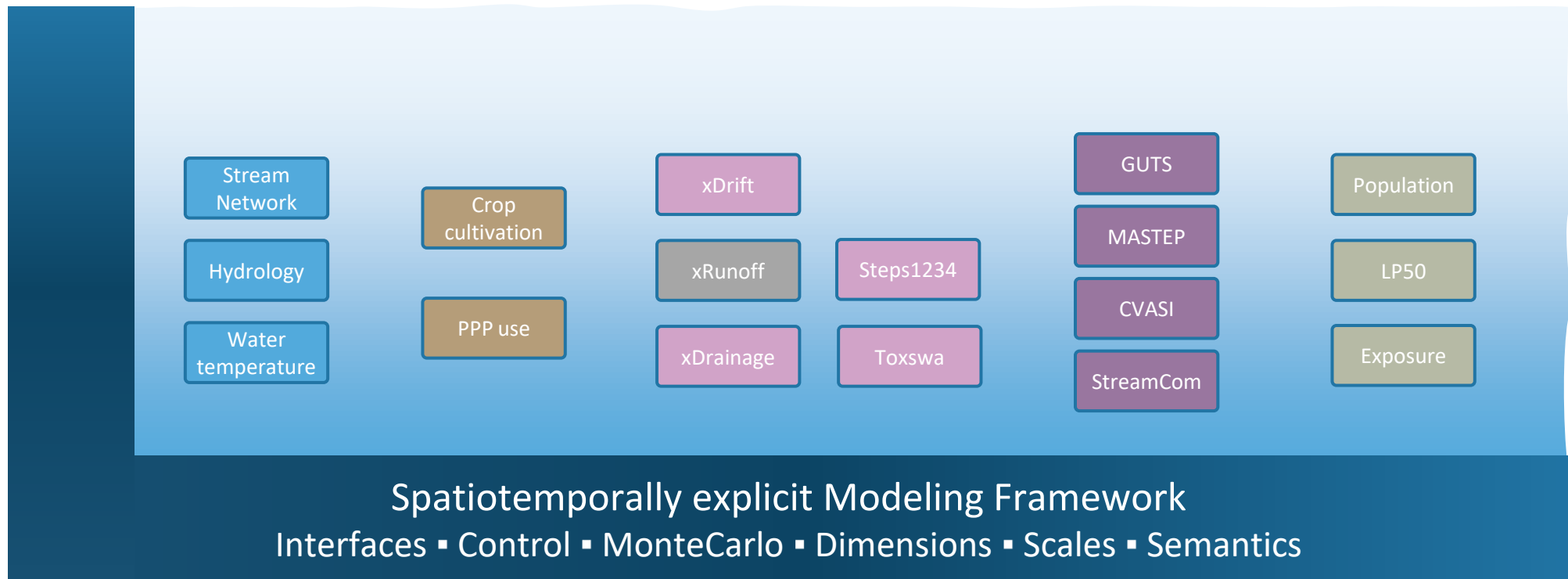
Technology – Multidimensional data arrays



<https://www.hdfgroup.org/> - Originally developed at the U.S. National Center for Supercomputing Applications, it is supported by The HDF Group, a non-profit corporation whose mission is to ensure continued development of HDF5 technologies and the continued accessibility of data stored in HDF

The Modelling Approach

xAquaticRisk

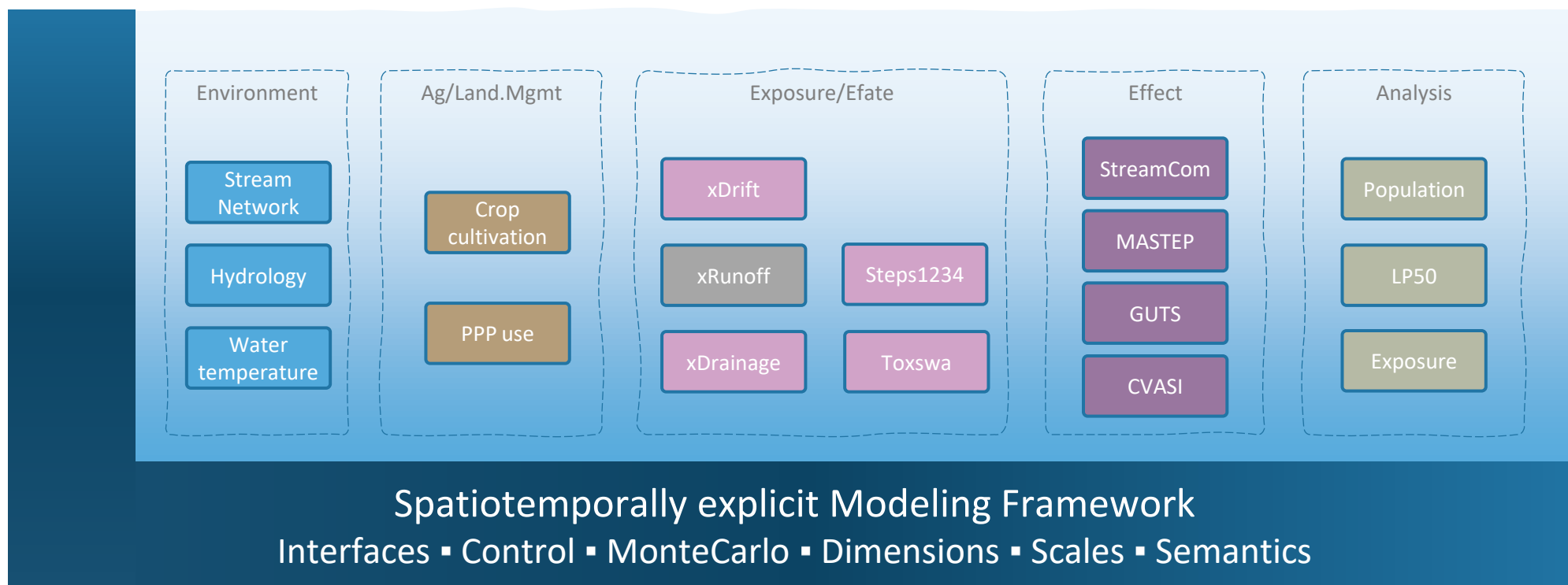


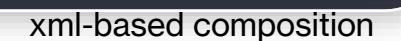


The Modelling Approach

xAquaticRisk

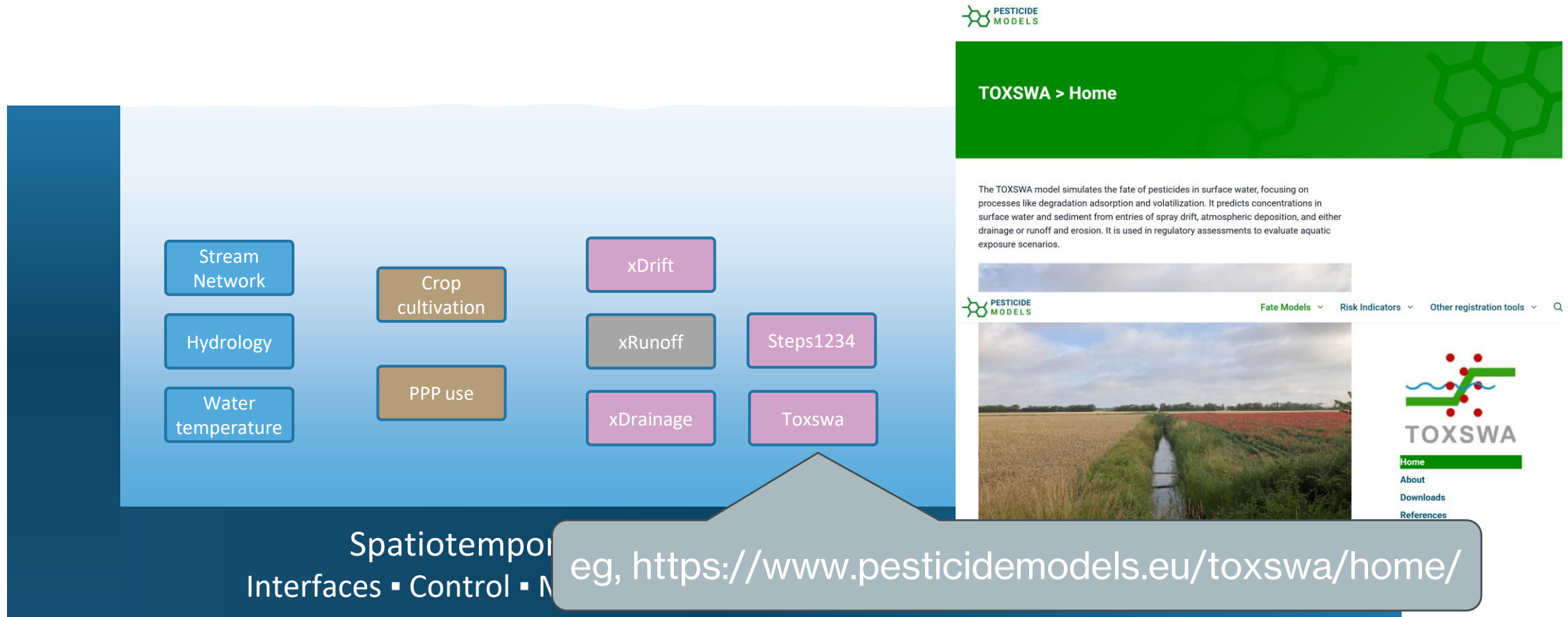
- ✓ **Modular**
- ✓ **Open Source, FAIR**
- ✓ **Makes use of regulatory established models**
- ✓ **Flexible, adaptable**





The Modelling Approach

xAquaticRisk/TOXSWA Component

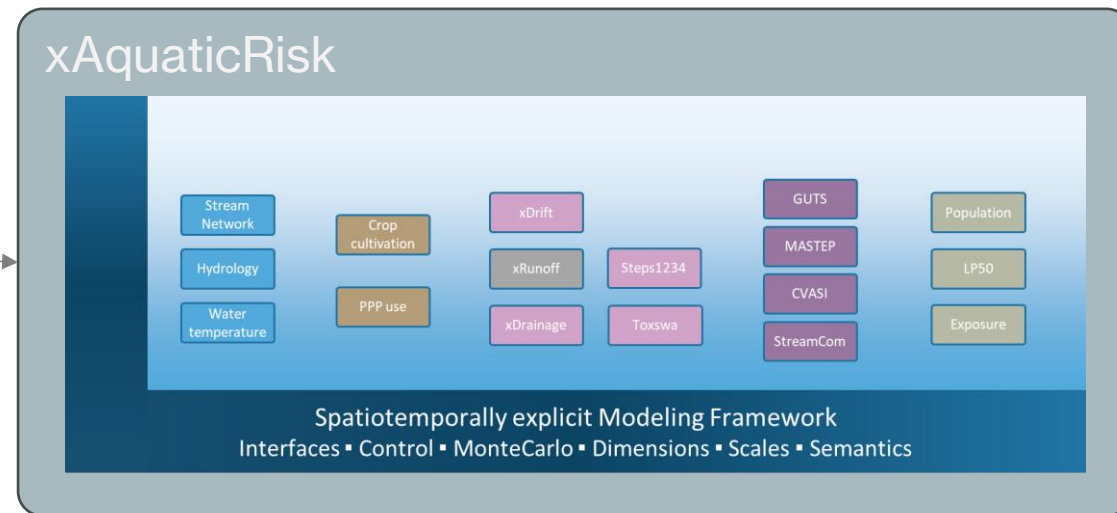


The Modelling Approach

xAquaticRisk



✓ User get in touch with a single model



xLandscape-based Model Examples



xLandscape - User Levels and Roles

User - API

- an xLandscape model as backend, called via API (Application Programming Interface)
- also: an xLandscape model as a module, embedded in other environments, eg, DigitalTwin

your role(s)
today and in
the future?!

User - Trained

- Trained xLandscape model user, with basic technical expertise, but within the need for expert domain knowledge
- Pre-prepared model parameterisation, execution and compilation of results
- In case, the role might include scenario development, requiring geospatial expertise

User - Expert

- Ready-to use xLandscape model application (eg, xAquatic), parameterisation (configuration) and execution
- Expert knowledge on xLandscape model domains (eg, xAquatic, xOffFieldSoil, xPollinator)
- In case, the role might include scenario development, requiring geospatial expertise

Landscape-Model Builder (eg, xAquaticRisk)

- Expert to build ready-to-use, specific landscape models from using the xLandscape core and components (eg, xOffFieldSoil, xAquaticRisk)
- Note: this role will particularly profit from AI-support

xLandscape Component Developer

- Component developer (integrating models/modules, eg, Toxswa, Guts, StreamCom, Mastep, PRZM, CropProtection)
- Expertise in specific model to be integrated and xLandscape component development (concepts, interface, python)

xLandscape Core Developer

- Core xLandscape developer. Conceptual development and implementation.
- Advanced expertise, eg, risk assessment domain, Informatics, semantics

Get xAquaticRisk

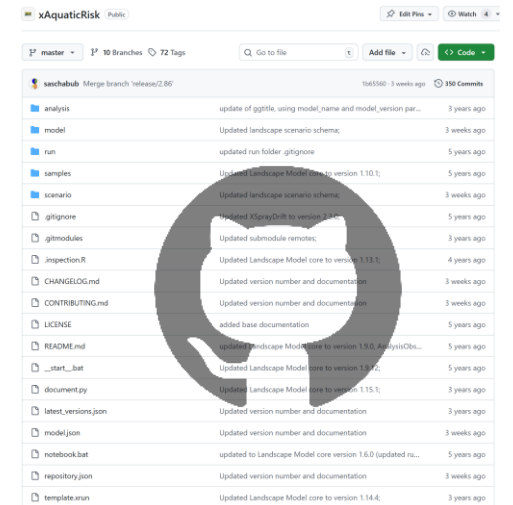
no installation needed = 'xcopy-ready'

download
xAquaticRisk.zip



or

clone from
Github





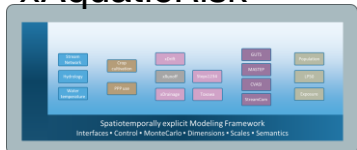
Use a **Web-Browser interface** if available for your model.

The screenshot shows the AWS IAM console interface. At the top, there are tabs for 'Users', 'Groups', 'Roles', and 'Sessions'. The 'Users' tab is selected. Below the tabs, there is a 'Users' section with a 'Users' link. The 'Users' list is empty. Below the 'Users' list, there is a 'Groups' section with a 'Groups' link. The 'Groups' list shows one group: 'AWSManagementConsoleA...'. The group has a 'Users' link. Below the 'Groups' list, there is a 'Roles' section with a 'Roles' link. The 'Roles' list is empty. Below the 'Roles' list, there is a 'Sessions' section with a 'Sessions' link. The 'Sessions' list is empty.

.xrun or .yaml input file

Model Configuration
-Expert Level-
.\model\variant\mc.xml

xAquaticRisk



Start a Simulation

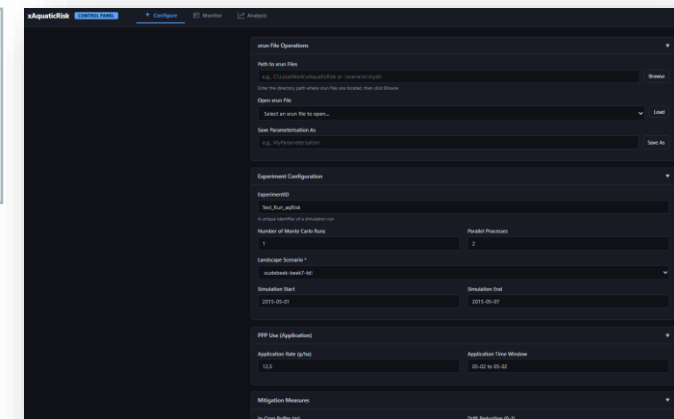
.git
 analysis
 model
 run
 samples
 scenario
 webui
 .gitignore
 .gitmodules
 .inspection.R
 __start__.bat
 bruchenbruecken_test_toxswa_2 - LGUTSOnly.xrun
 bruchenbruecken_test_toxswa_2 - LGUTSOnly_Mitig.xrun
 bruchenbruecken_test_toxswa_2.xrun
 bruchenbruecken_test_toxswa_2b.xrun

Option (1): **Windows Explorer**
 In your xAquaticRisk folder,
drag-and-drop your xrun (yaml) file
 onto the **__start__.bat**

Option (2): **CMD Window**
 In your Command Window,
 call **__start__.bat + your xrun (yaml) file**

```
C:\LocalWork\xAquaticRisk>__start__.bat template.xrun|
```

Option (3):
Web-browser
User Interface



'ControlPanel'

Advanced - your IDE as Operations Center

Visual Studio Code

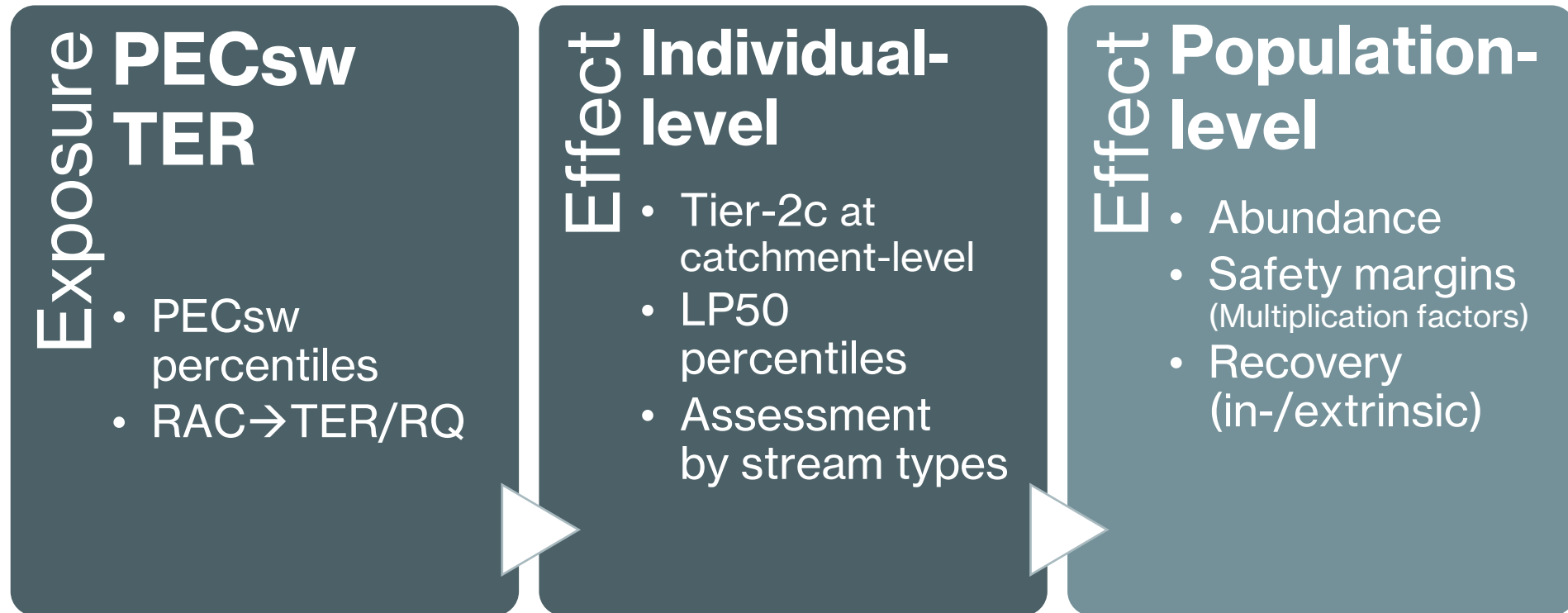
- Development, Github
 - Model input preparation
 - Model execution
 - Analysis (eg, Jupyter Notebooks)
 - Documentation
- ..all AI-suported (Copilot)

The screenshot displays the Visual Studio Code IDE with the xAQuaticRisk project open. The interface is divided into several panels:

- Source Control:** Shows a list of repositories (xAQuaticRisk, AnalysisObserver, CascadeToxswa, CmfContinuous, LEEffectModel, LP50, ReportingObserver, StreamCom, XSprayDrift, core) and a list of changes (fix(xcopy): revert XAQ_RUN_DIR, merge(feature): feature/integrate-an..., fix(explorer): resolve scenario path roots and define S..., feat(explorer): add run-folder, experiment, and MC sele..., docs: add code-fence languages in startup/configuration..., docs: update startup defaults and clarify setup_all.bat is..., feat(analysis): derive default scenario root by replacing t..., feat(startup): default run root to C:\ when XAQ_RUN_DI..., fix(ui): use Windows separator for default scenario folde..., refactor(analysis): default scenario folder to run_root/sc..., fix(startup): make start.bat Explorer-friendly with run-dir..., feat(analysis): add configurable scenario folder with def..., fix(ui): show only Analysis and Explorer in xAQuaticRiskA..., feat(xcopy): make xAQuaticRiskAnalysis self-contained..., fix(start.bat): correct delayed expansion syntax for pyth..., feat: integrate analysis payload and add setup/validation...).
- Main Editor:** Displays the 'Parameterisation' tab, showing the 'Parameterisation (xrun-) File Operations' section. It includes fields for 'Path to Parameterisation Files (xrun / .yaml)', 'Open Parameterisation File', 'Save Parameterisation As', and 'Experiment Basic Settings' (ExperimentID, Number of Monte Carlo Runs, Parallel Processes, Landscape Scenario, Simulation Start, Simulation End, Plant Protection Product Use (PPP Application), Application Rate (g/ha), Application Time Window).
- Chat:** Shows Copilot suggestions for fixing issues. It lists 'FAIL fixes applied' and 'WARN fixes applied' with specific actions like 'Removed the silent default', 'Pinned geopandas==1.0.3 and pyogrio==0.10.0', 'Added analyze_server.py', 'Corrected "Python 3.9" to "Python 3.12" to match what', and 'Actually downloads'.
- Terminal:** Shows the output of the application, including browser connections and warnings.

xAquaticRisk

Model Outcome, Modelling Endpoints



let's start an
analysis

Analysis & Reporting

Jupyter Notebooks, WebUI

Jupyter Notebooks* (Python)
→ tables (csv, excel), graphs, maps, movies
location: xAquaticRisk\analysis

xAquatic - Exposure and Effect Basic Analysis

(Thorsten Schäd, version 2025.0.1.0)

This notebook provides basic analysis for exposure (PECsw and PECsed) and effect (individual level, GUTS, survival, LP50) characterisation for xAquatic experiments.

- percentile tables (csv, excel)
- plots: time series, box plots, contour, maps

The analysis currently focuses on a single Monte Carlo run.

Python preparation

```
1 %matplotlib widget
2 #matplotlib inline
3
4 > import ...
18 #from matplotlib_scalebar.scalebar import ScaleBar
19 #from cmath import log10
20 #import hvplot.pandas # registers hvplot on pandas
21 #from scipy.interpolate import griddata
[1]
```

User Input

Select Monte Carlo run

```
1 # !the outcome folder structure of xAquatic is expected: ...\\run\\Experiment-ID\\mcs\\MC-ID\\store\\arr.dat
2 # path to Monte Carlo run to be analysed (eg, copied from explorer, eg, C:\\LocalWork\\xAquaticRisk\\run\\rummen_toxswa_10yrs_lateMitigati
3 path = r'E:\\xAquaticRisk\\run\\bruchenbruecken_toxswa_10yrs_noMitig_LGUTSonly\\mcs\\X3940E4U0XSCQV0ISS'
4 #parking lot
5 # E:\\xAquatic_Experiments_Store\\run\\funne_arableApril1x_Miti_1\\mcs\\X3HQ6CS4A06FTWF6V
6 # E:\\xAquatic_Experiments_Store\\run\\rummen_toxswa_10yrs_1\\mcs\\X3HEF7VESLXT6FRFAV
7 # E:\\xAquatic_Experiments_Store\\run\\rummen_toxswa_10yrs_earlyMitigation_3\\mcs\\X3ESL54CYNDVEIHEC1
8 # path to related scenario (eg, C:\\LocalWork\\xAquaticRisk\\scenario\\Scenario-Rummen)
```

functions

iFEM

lib

reporting

sample

ARE_LP50.ipynb

Default Reporting Elements.ipynb

default_reporting_elements.R

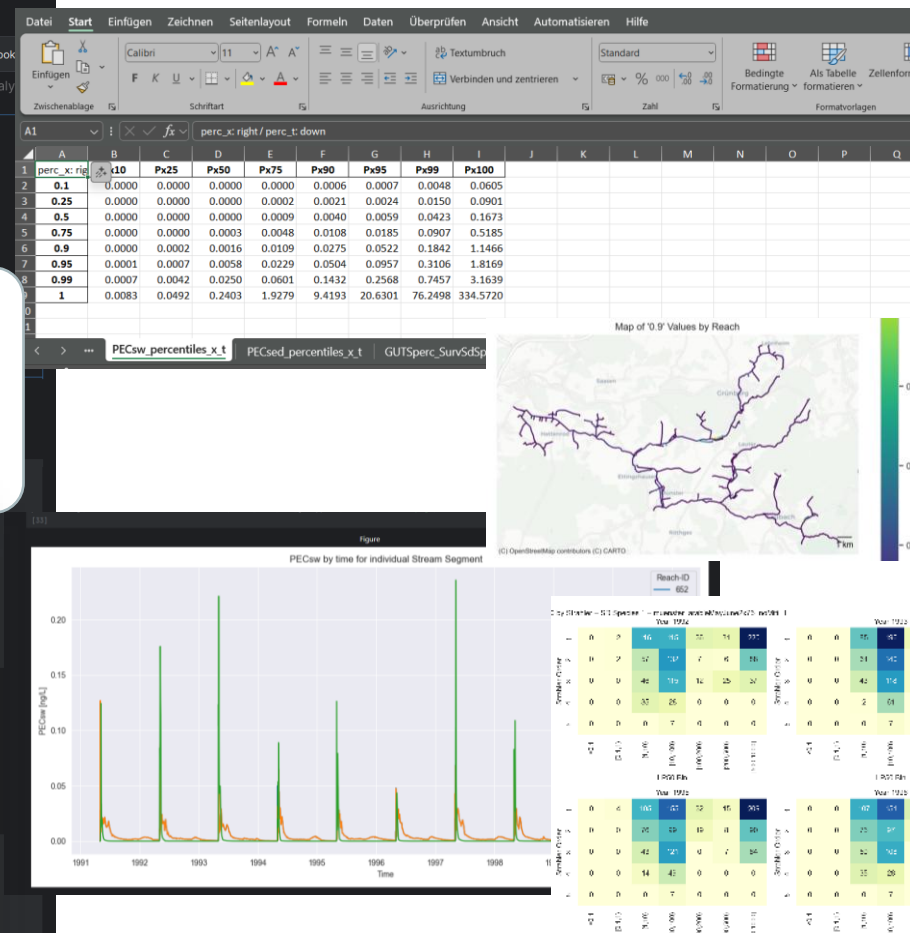
Example Figure StreamCom.ipynb

Hydrographic Maps.ipynb

LP50 by Strahler Order.ipynb

LP50 Distribution.ipynb

PECsw_toExcel.ipynb

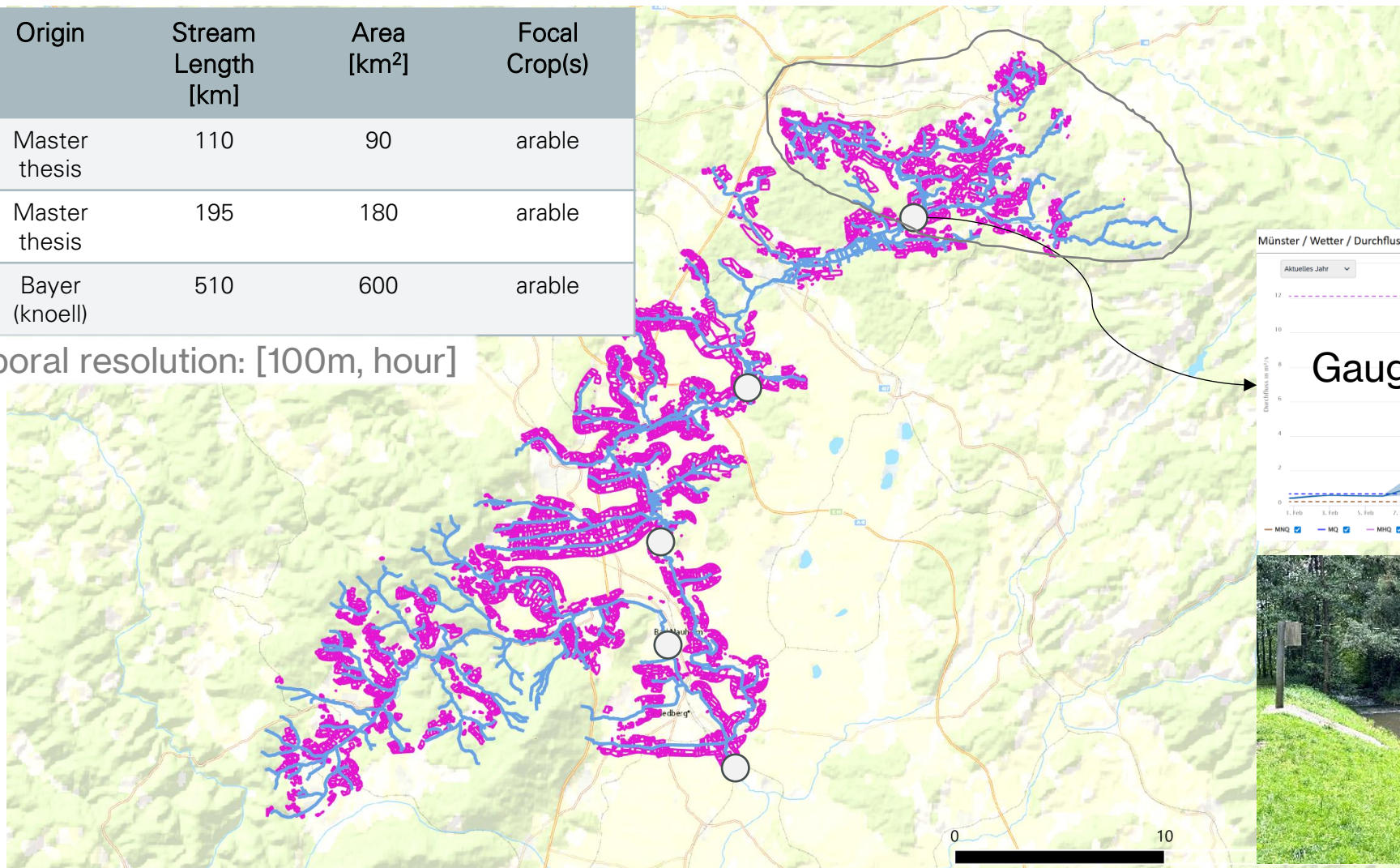
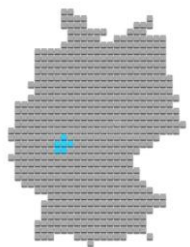


*experience made: discarded eg, MS Power BI and others for accessibility, simplicity and processing speed

Scenarios – Wetterau/Münster

Sub-Catchment	Origin	Stream Length [km]	Area [km ²]	Focal Crop(s)
Muenster	Master thesis	110	90	arable
Muschenheim	Master thesis	195	180	arable
Bruchenbrücken	Bayer (knoell)	510	600	arable

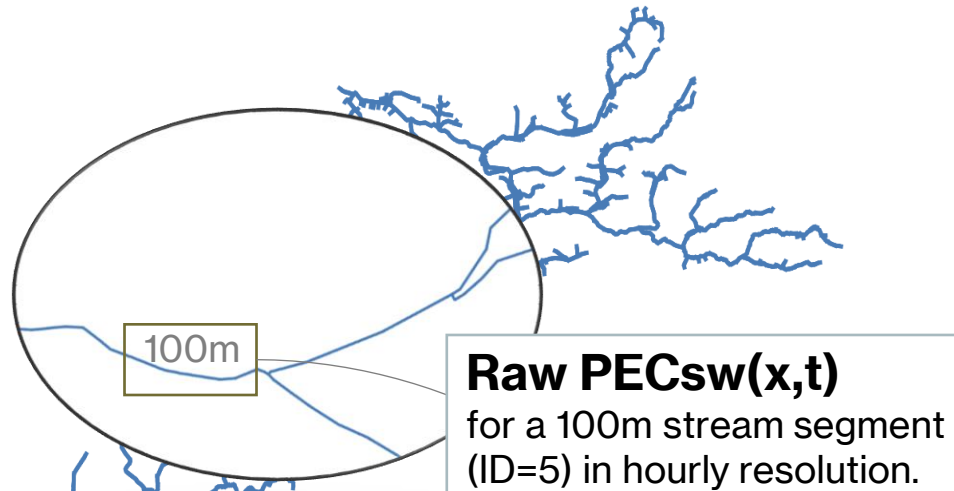
Spatial & temporal resolution: [100m, hour]



<https://github.com/xlandscape/Scenario-Muenster> (work in progress)



Spatiotemporally Explicit Aquatic Exposure



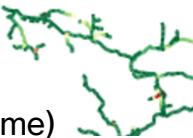
	0	1	2	3	4	5	6	7	8	9	10
428490.0	0.0	7.3324E...	1.90788...	3.88988...	1.92217...	1.77108...	2.72551...	2.802E-12	6.29665...	8.8994...	
428500.0	0.0	7.33038...	1.90711...	3.88798...	1.92133...	1.77044...	2.72465...	2.80050...	6.29285...	8.8970...	
428510.0	0.0	7.31496...	1.90631...	3.88656...	1.92057...	1.76995...	2.7334E...	2.79954...	6.28993...	8.9254...	
428520.0	0.0	7.31171...	1.90544...	3.88511...	1.9267E...	1.77585...	2.73141...	2.79902...	6.28763...	8.91431...	
428530.0	0.0	7.30595...	1.90457...	3.88325...	1.92490...	1.77455...	2.73019...	2.80885...	6.30942...	8.9084...	
428540.0	0.0	7.2972E...	1.90369...	3.88172...	1.92337...	1.77339...	2.7291E...	2.80655...	6.30436...	8.9072...	
428550.0	0.0	7.28954...	1.90272...	3.88009...	1.92204...	1.77246...	2.72825...	2.8048E...	6.30043...	8.9056...	
428560.0	0.0	7.28241...	1.90178...	3.87813...	1.92087...	1.77175...	2.72745...	2.80348...	6.29751...	8.9035...	
428570.0	0.0	7.27105...	1.9009E...	3.87624...	1.91986...	1.77109...	2.72679...	2.80251...	6.29503...	8.9021...	
428580.0	0.0	7.26428...	1.90003...	3.87469...	1.91896...	1.77057...	2.72633...	2.80180...	6.29293...	8.9011...	
428590.0	0.0	7.26001...	1.89915...	3.87288...	1.91816...	1.7702E...	2.72592...	2.80118...	6.29163...	8.9003...	
428600.0	0.0	7.25501...	1.89828...	3.87101...	1.91743...	1.76983...	2.7256E...	2.80056...	6.2911E...	8.8988...	
428610.0	0.0	7.24541...	1.89731...	3.86946...	1.91668...	1.76945...	2.72542...	2.80014...	6.29056...	8.9285...	
428620.0	0.0	7.23363...	1.89637...	3.86788...	1.91586...	1.76918...	2.72508...	2.79987...	6.29006...	8.9185...	
428630.0	0.0	7.23110...	1.89549...	3.86569...	1.91510...	1.76900...	2.73429...	2.79959...	6.28973...	8.9113...	
428640.0	0.0	7.22791...	1.89462...	3.86395...	1.91447...	1.7688E...	2.7323E...	2.79925...	6.28926...	8.9064...	
428650.0	0.0	7.21685...	1.89365...	3.86243...	1.91387...	1.76857...	2.73059...	2.79906...	6.28839...	8.9033...	
428660.0	0.0	7.22207...	1.89240...	3.86054...	1.91331...	1.76830...	2.72926...	2.79883...	6.2878E...	8.9008...	
428670.0	0.0	7.21531...	1.89103...	3.85772...	1.91270...	1.76804...	2.72825...	2.79838...	6.28728...	8.8987...	
428680.0	0.0	7.20563...	1.88963...	3.85513...	1.91201...	1.76775...	2.71782...	2.79792...	6.28654...	8.8658...	
428690.0	0.0	7.20511...	1.88813...	3.85258...	1.911290...	1.76745...	2.7188E...	2.79747...	6.28569...	8.8728...	
428700.0	0.0	7.19776...	1.88658...	3.84991...	1.91051...	1.76705...	2.71944...	2.79686...	6.28451...	8.8777...	
428710.0	0.0	7.19091...	1.88501...	3.84652...	1.90971...	1.76654...	2.71977...	2.79611...	6.28305...	8.8813...	
428720.0	0.0	7.18573...	1.88343...	3.84343...	1.90886...	1.76606...	2.71992...	2.79525...	6.28137...	8.8829...	
428730.0	0.0	7.18058...	1.88681...	3.8339E...	1.90796...	1.7655E...	2.72000...	2.79431...	6.27949...	8.8833...	
428740.0	0.0	7.17125...	1.87893...	3.83098...	1.90691...	1.76487...	2.7199E...	2.7933E...	6.27744...	8.8826...	

→ Calculation of **PECsw Percentiles**
in time and space in 2 steps:

(1) PECsw percentile **over time**
for each stream segment

→ spatial distribution of PECsw(x)_pt
→ a map of local PECsw percentiles over time

eg, PECsw(x)_pt75 (75th percentile over time)



(2) PECsw percentile **over space**
from PECsw(x)_pt

→ **single value PECsw_pxpt**

eg, PECsw_px90pt90 = 1.9 ng/L

→ to be used in Risk Assessment

Analysis - Exposure

PECsw percentiles over time and space, **PECsw_pxpt**

PECsw Percentile Table [ng/L] — rows: temporal percentiles, columns: spatial percentiles ▼ hide

pt ↓ / px →	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
Pt10	0	0	0	0	0	0	0	1.00e-4
Pt25	0	0	0	0	0	0	0	4.00e-4
Pt50	0	0	0	0	1.00e-4	1.00e-4	1.00e-4	0.0027
Pt75	0	0	0	2.00e-4	5.00e-4	6.00e-4	9.00e-4	0.0156
Pt90	0	0	1.00e-4	0.0014	0.0039	0.0048	0.0077	0.0851
Pt95	0	0	5.00e-4	0.0042	0.0135	0.0151	0.0209	0.1969
Pt99	0	0	0.0050	0.0422	0.0796	0.1153	0.1747	0.7717
Pt100	0	0	1.0285	4.8364	18.4861	39.5371	133.5139	314.5960

1 Monte Carlo run. Analysis in Jupyter Notebook or using the 'ControlPanel'

Analysis - Exposure

95 percent of stream segments have a PEC_{sw} < 0.0048 ng/L, in 90 percent of time

→ Modelling Endpoint definition guided by SPGs and to be set in relation to RAC

PEC_{sw} Percentile Table [ng/L] — rows: temporal percentiles, columns: spatial percentiles ▼ hide

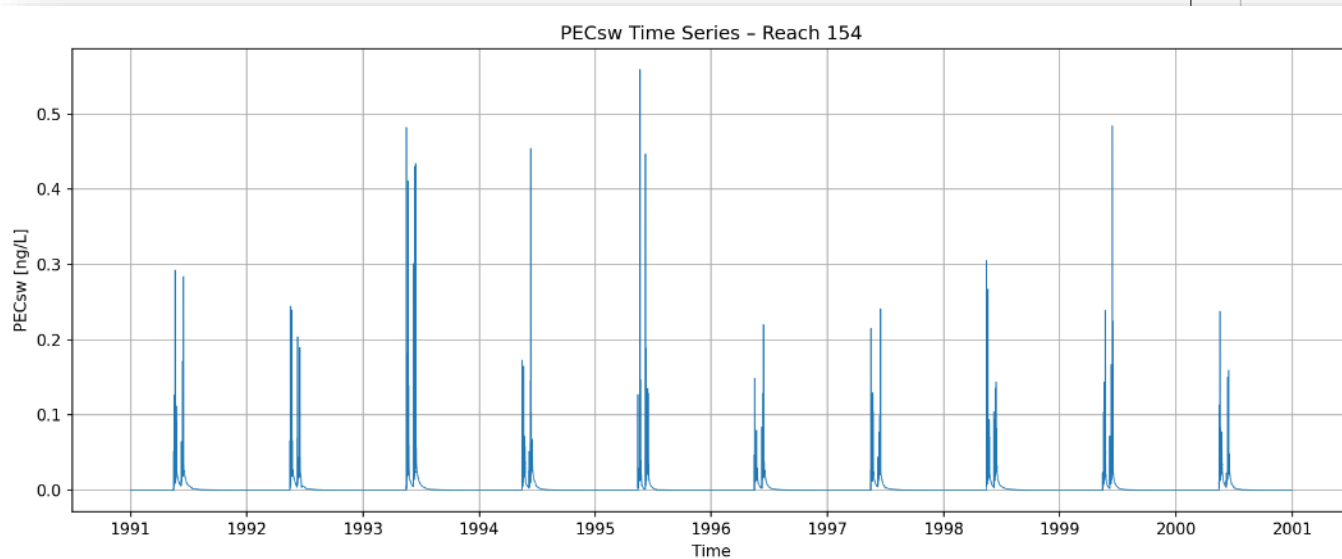
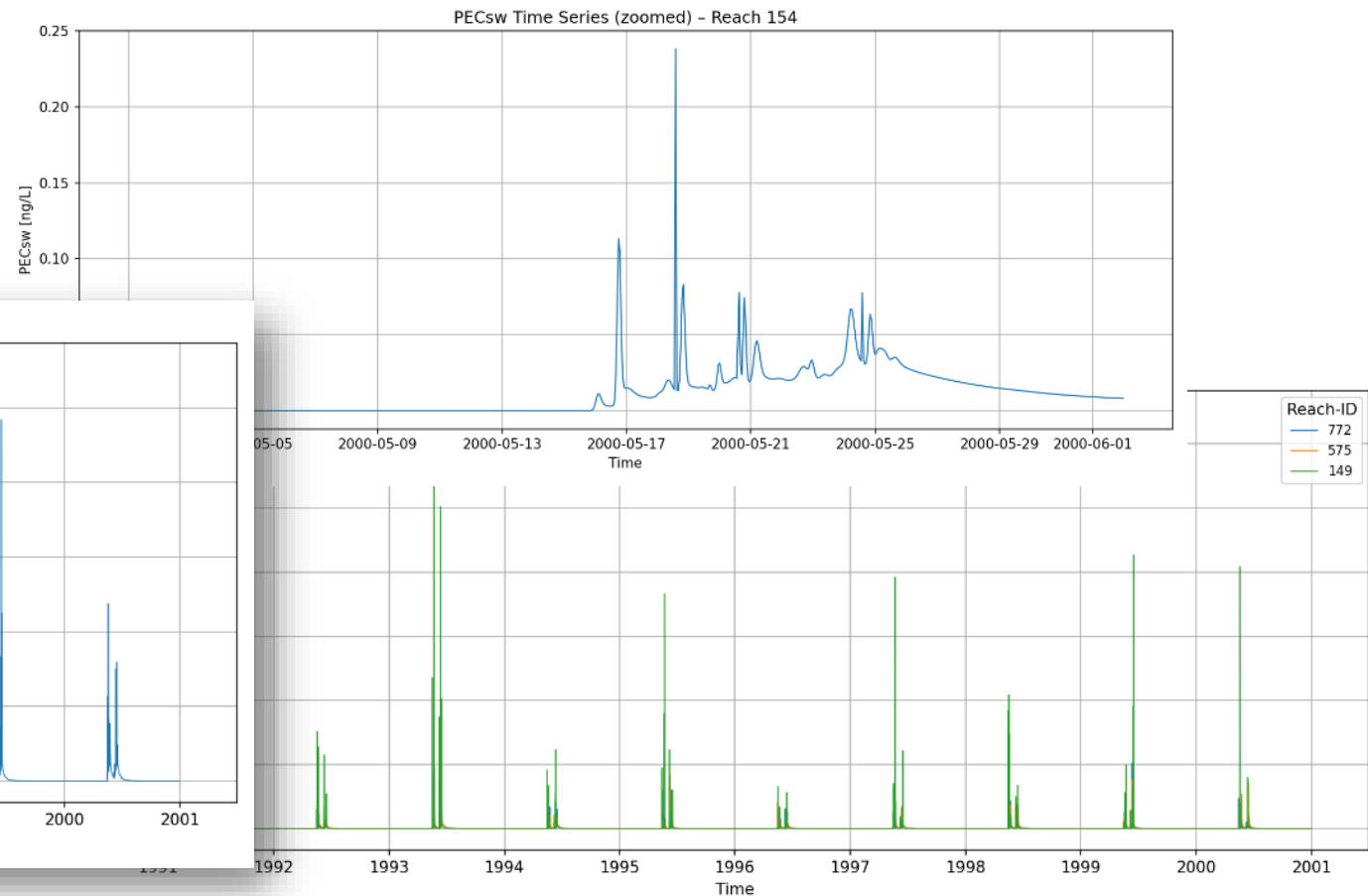
pt ↓ / px →	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
Pt10	0	0	0	0	0	0	0	1.00e-4
Pt25	0	0	0	0	0	0	0	4.00e-4
Pt50	0	0	0	0	1.00e-4	1.00e-4	1.00e-4	0.0027
Pt75	0	0	0	2.00e-4	5.00e-4	6.00e-4	9.00e-4	0.0156
Pt90	0	0	1.00e-4	0.0014	0.0039	0.0048	0.0077	0.0851
Pt95	0	0	5.00e-4	0.0042	0.0135	0.0151	0.0209	0.1969
Pt99	0	0	0.0050	0.0422	0.0796	0.1153	0.1747	0.7717
Pt100	0	0	1.0285	4.8364	18.4861	39.5371	133.5139	314.5960

full picture: informative PEC_{sw}_pxpt context around selected exposure modelling endpoint

Analysis - Exposure

PECsw/sed(x,t) Time Series

→ Insights on system dynamics behaviour
Verification. Supportive information.
Research, knowledge gain.
Outlook: Monitoring design.



Analysis - Exposure

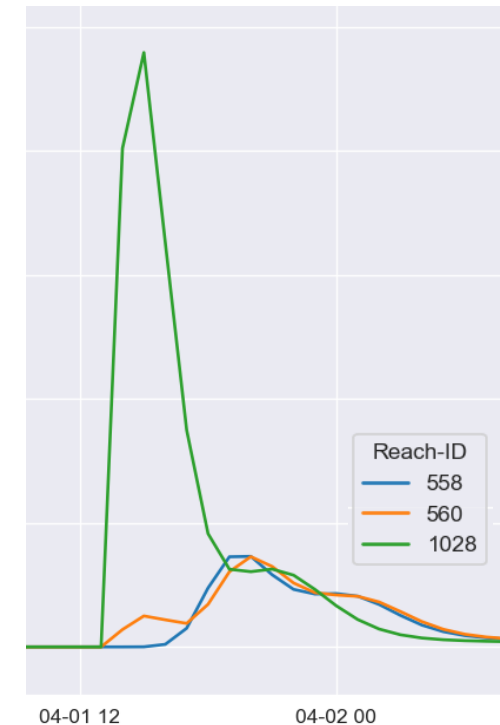
PEC_{sw}(x,t) - Time Series at confluences

→ Insights on system dynamics behaviour.

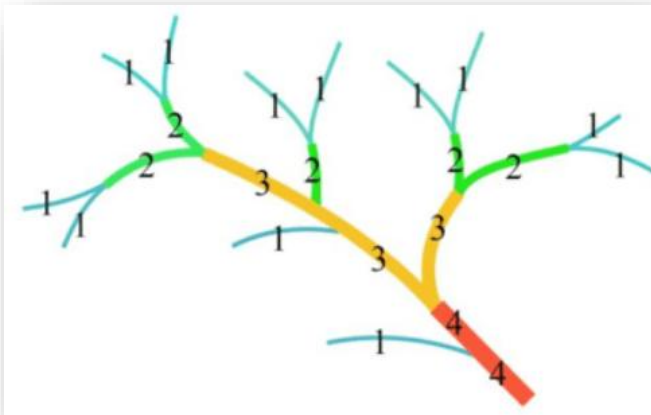
Knowledge gain. Verification.

Supportive information.

Monitoring design.

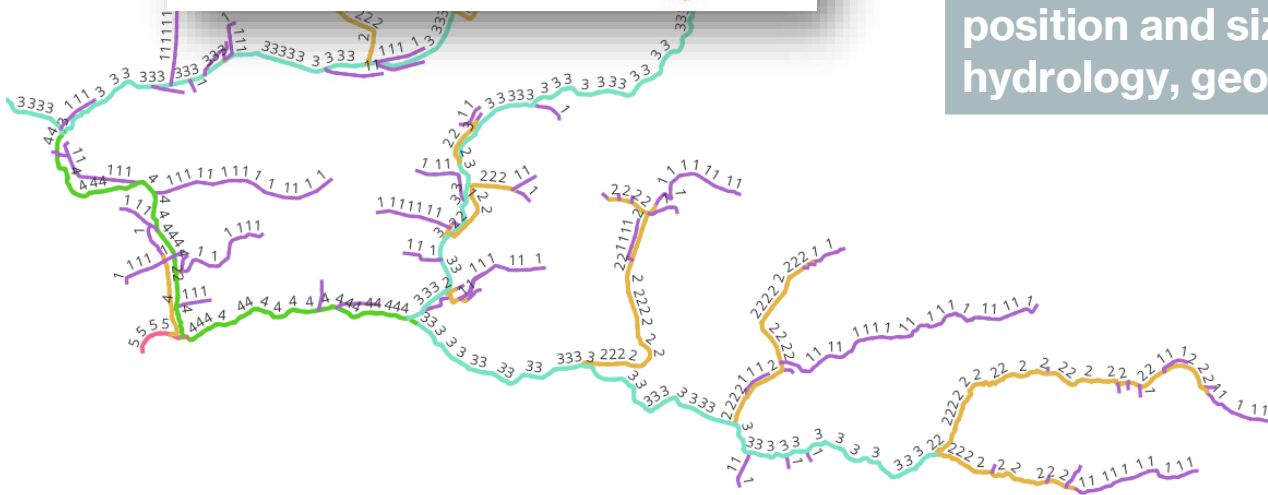


Analysis - Exposure by Stream Types



Strahler order is a hierarchical way to classify streams based on how tributaries join.

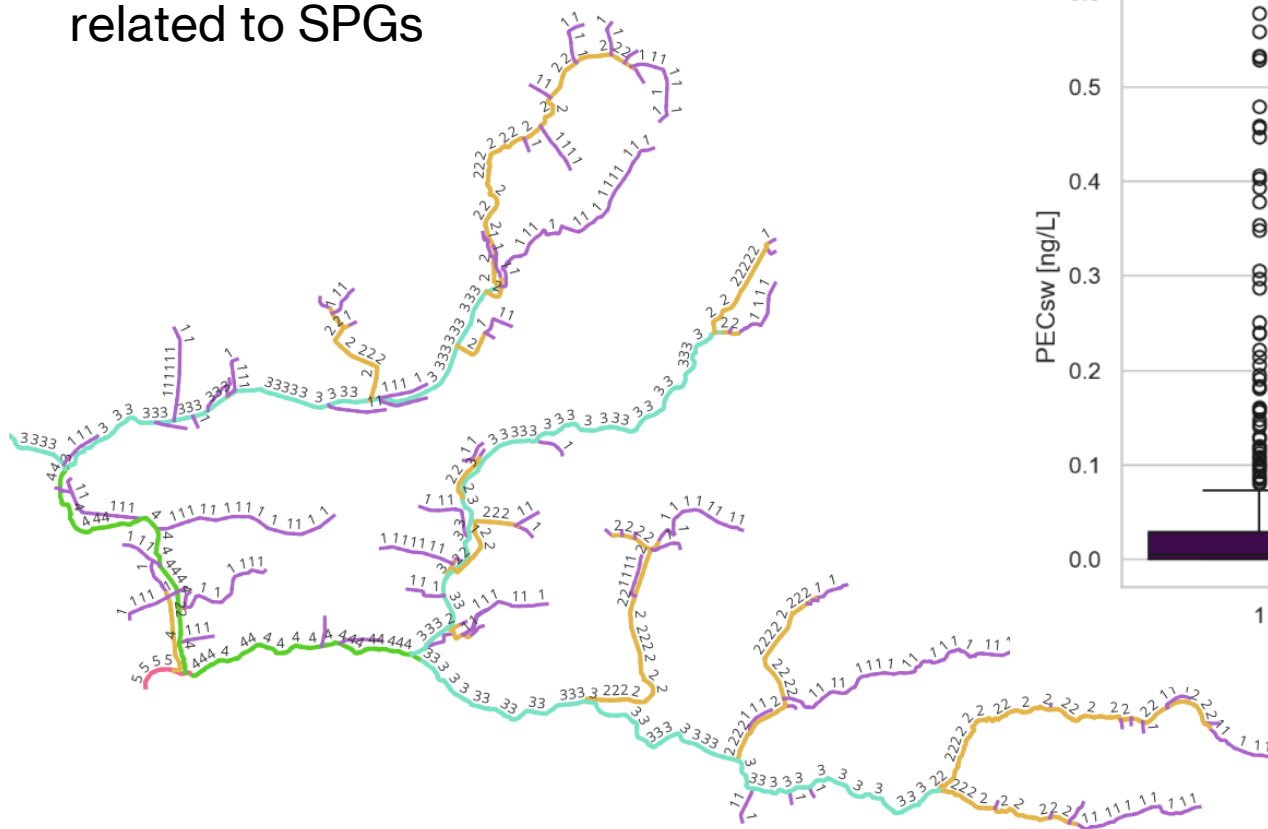
The smallest, unbranched channels are 1st-order streams. **When two streams of the same order meet, the order increases by one** (e.g., two 1st-order streams form a 2nd-order stream). When streams of different orders meet, the downstream segment keeps the higher order. It provides a simple, standardized measure of **a stream's relative position and size** within a drainage network, often **used to describe hydrology, geomorphology, and ecological patterns**.



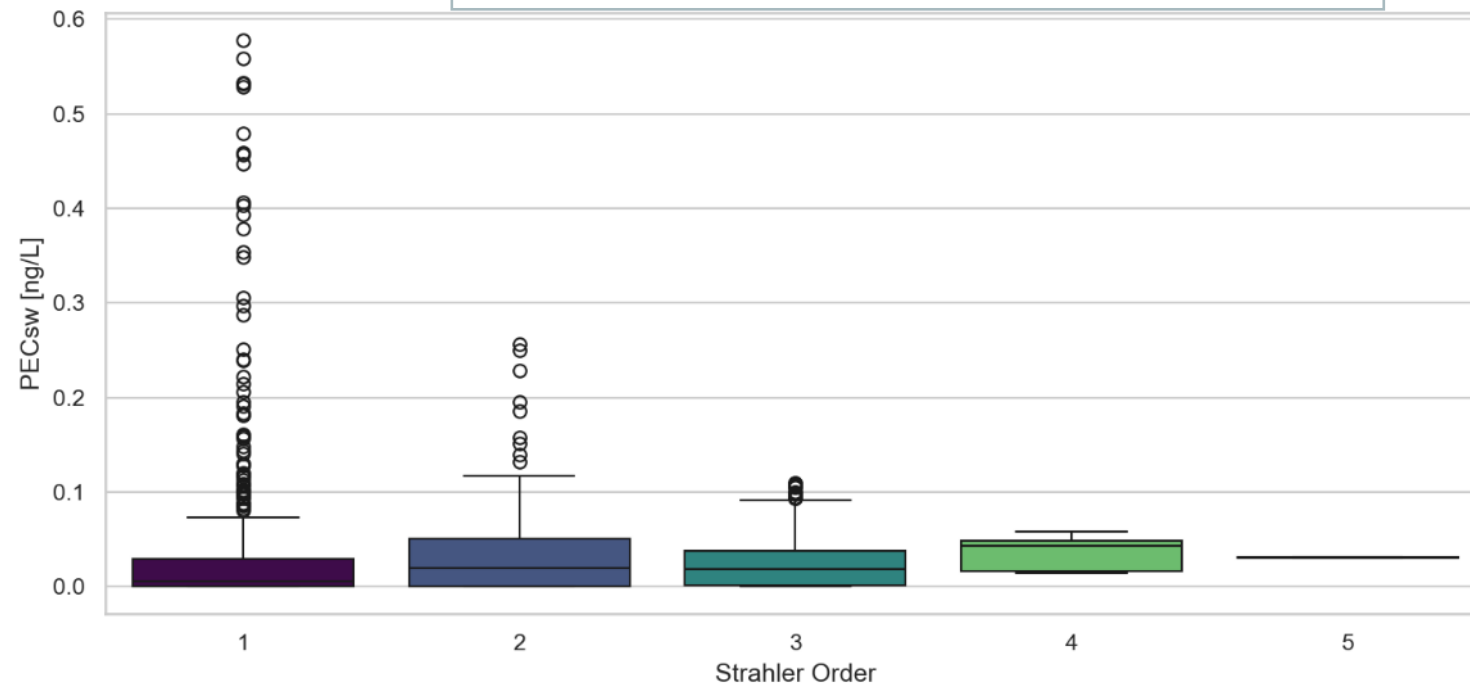
Analysis - Exposure by Stream Types

Distribution of PECsw(x)_pt by Stream Order

→ Evaluation by **ecological conditions** related to SPGs



PECsw(x)_pt99 [ng/L] (by stream Strahler order)



Analysis - Exposure

Geospatial Visualisation and Analysis

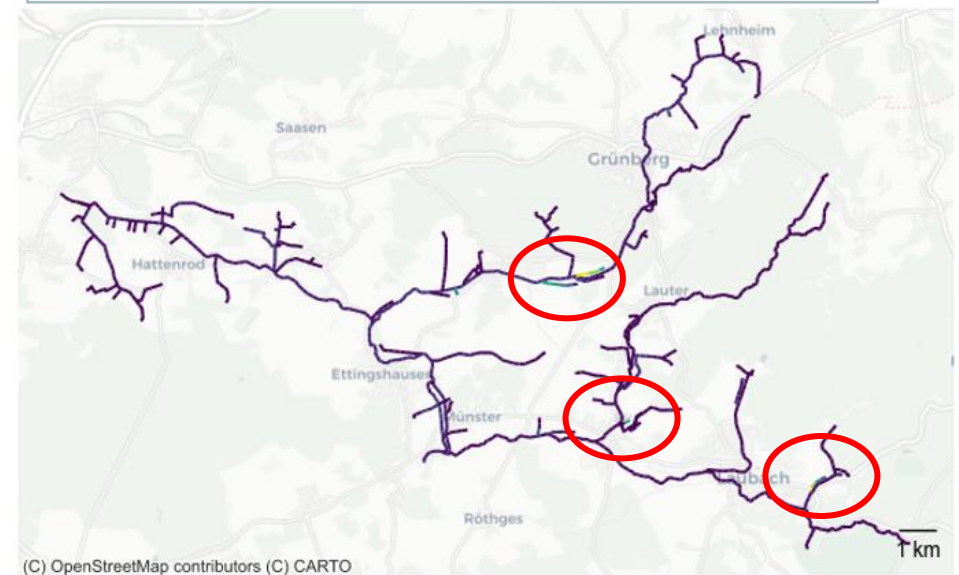
Maps illustrate the spatial distribution of endpoints.
→ Hot-Spot analysis. Recovery. Transparency. Trust.

Knowledge gain: Mapping supports **what-if analysis**, eg to show the effect of **risk management** to **hot-spots** distribution.
→ derive general rules for catchment management

Geodata verification and refinement: in the Münster catchment, a **field visit** has indicated that the current mapping is conservative, eg,

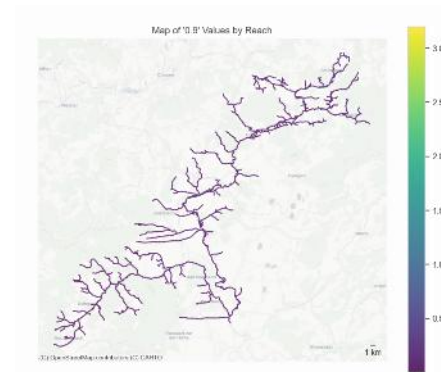
- occurrence of arable fields and their distance to streams
- many ditches contain water only temporarily (→ species occurrence)
- presents of riparian vegetation as drift filter

PECsw(x)_pt90 [ng/L]
(spatial distribution of local 90th percentile over time)



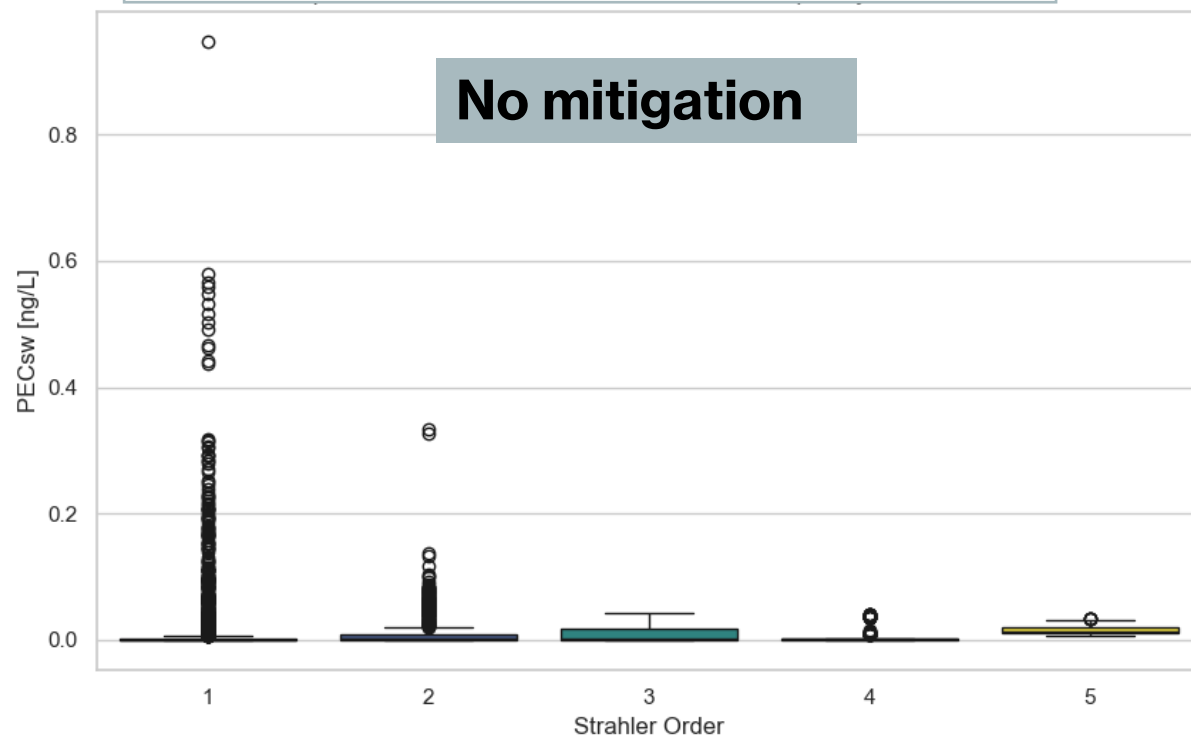
of particular interest for
aquatic risk in SEU

Analysis - Exposure Mitigation



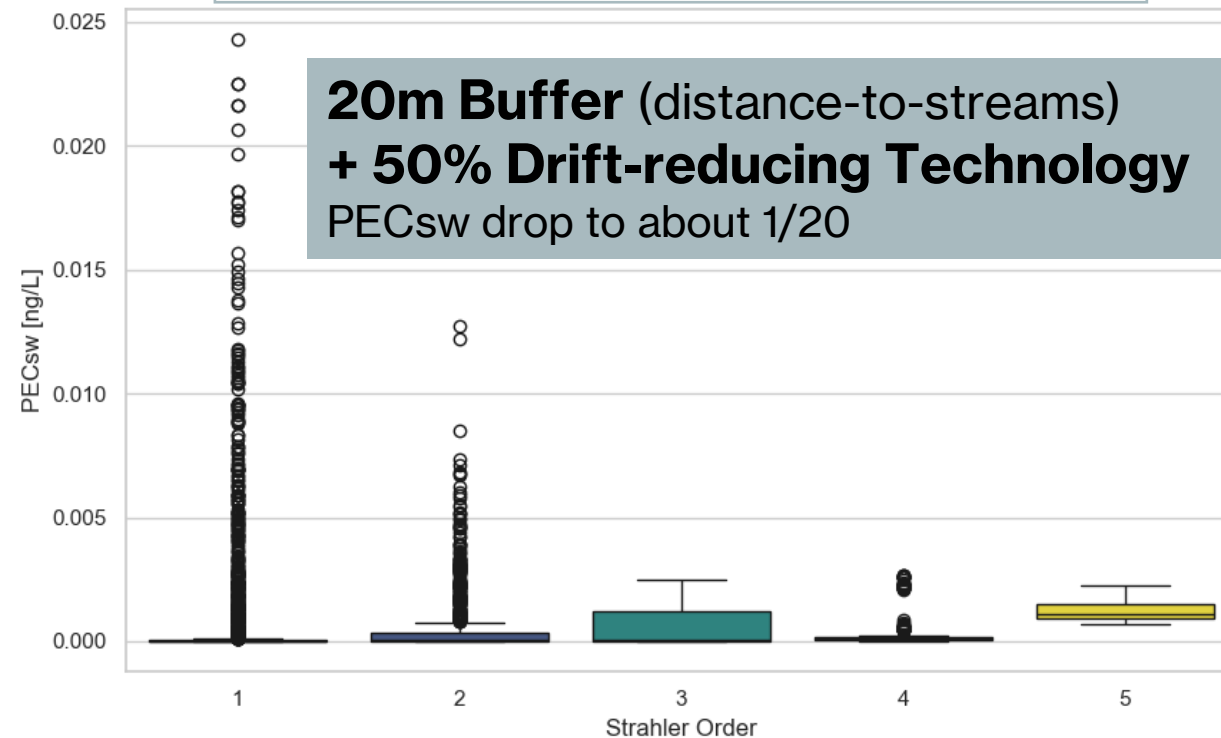
PECsw(x)_pt90 [ng/L] (by stream Strahler order)

No mitigation



PECsw(x)_pt90 [ng/L] (by stream Strahler order)

**20m Buffer (distance-to-streams)
+ 50% Drift-reducing Technology**
PECsw drop to about 1/20



Supplement

Risk Mitigation and Management

FOCUS Landscape & Mitigation (2008)

SETAC MAgPIE Workshop – Mitigating the Risks of Plant Protection Products in the Environment

A two-part workshop (Rome 2013, Madrid 2013), organised under the auspices of SETAC and the European Commission, gathering 95 experts from 24 European countries (regulators, industry, academia, extension services).

Objective: Inventory and harmonise risk mitigation measures (RMMs) for pesticides across Europe, under EU Regulation (EC) No 1107/2009, to reduce environmental transfers to non-target compartments.

Key Outcome – Mitigation Toolbox: A harmonised European toolbox of RMMs ranked by: (1) scientific evidence of effectiveness, (2) practicality to farmers, (3) ease of regulatory implementation, (4) integration into risk assessment. Core categories include **direct measures** (spray-drift reduction technology at source) and **indirect measures** (no-spray zones, buffer strips to protect sensitive environments).

Relevance: MAgPIE-recommended measures (e.g., buffer distances, drift-reducing nozzles) align directly with the mitigation scenarios modelled in xAquaticRisk (see slides on exposure mitigation analysis).

Alix, A., Brown, C., Capri, E., Goerlitz, G., Golla, B., Knauer, K., Laabs, V., Mackay, N., Marchis, A., Poulsen, V., Alonso Prados, E., Reinert, W., Streloke, M. (Eds.) (2017). Mitigating the Risks of Plant Protection Products in the Environment: MAgPIE. SETAC Press. ISBN 978-1-880611-99-9. <https://www.setac.org/magpie>

Analysis - Effects

GUTS, Introduction

CaseStudy:

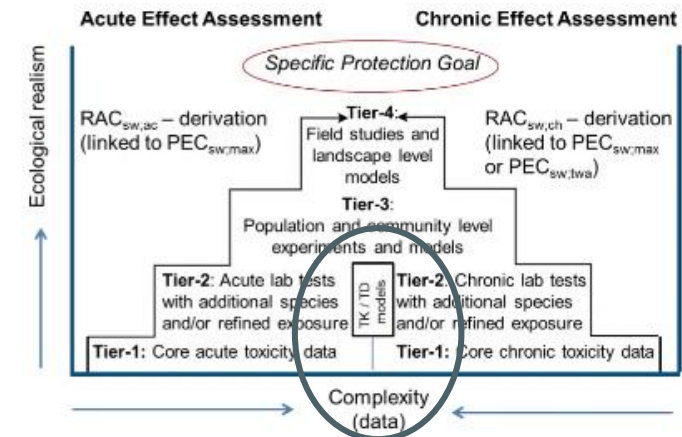
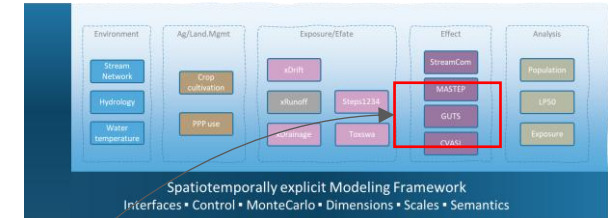
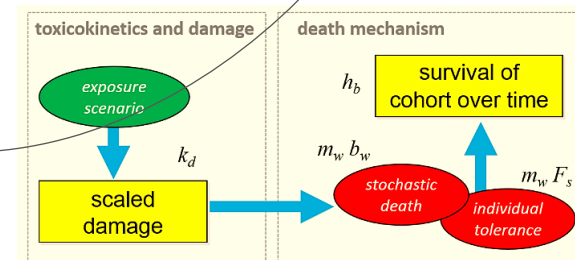
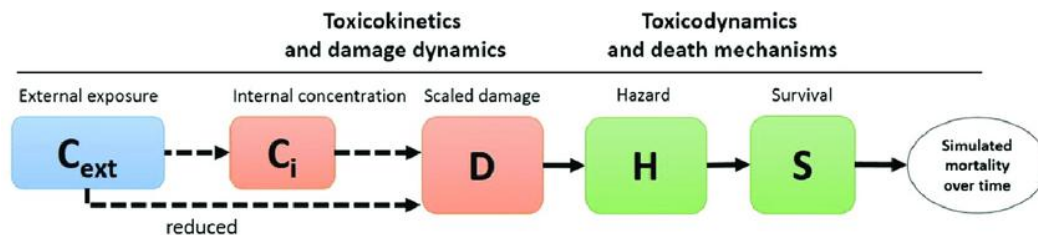
3 Species: Asellus aquaticus (Sp1), Cloeon dipterum (Sp2), Gammarus pulex (Sp3)

GUTS Models: IT = Individual Tolerance, SD = Stochastic Death

General Unified Threshold Models of Survival (GUTS)

are mechanistic toxicokinetic–toxicodynamic (TKTD) models used to describe and **predict the survival of aquatic invertebrates under time-variable chemical exposure**.

GUTS links external exposure to internal damage processes and organism-level mortality, allowing consistent interpretation of toxicity data across different exposure scenarios. By accounting for exposure dynamics, GUTS provides a scientifically robust approach for risk assessment beyond traditional static toxicity endpoints.



Supplement

General Unified Threshold Models of Survival (GUTS)

GUTS–IT: Individual Tolerance

Individuals differ in their tolerance to toxicant. Each individual has its own fixed internal threshold for damage. Death occurs immediately and deterministically when that individual threshold is exceeded.

The population shows gradual mortality because thresholds are distributed across individuals (e.g. log-logistic distribution). Once an individual's threshold is crossed, death is certain (no randomness).

Key characteristics: Deterministic at the individual level. Variability is between individuals, not in the death process. No recovery once the threshold is exceeded.

Conceptual interpretation: “Who dies depends on how tolerant the individual is.”

Typical implications: Mortality often occurs in steps when exposure increases. Particularly applicable for chemicals where sensitivity clearly varies among individuals

GUTS–SD: Stochastic Death

All individuals share the same internal threshold for damage. Exceeding the threshold does not cause immediate death. Instead, it increases the probability of dying per unit time. How mortality emerges: Death is a random (stochastic) process at the individual level. Higher damage → higher instantaneous death rate.

Key characteristics: Probabilistic at the individual level: Variability is in the timing of death, not in tolerance.

Individuals may survive some time even after exceeding the threshold.

Conceptual interpretation: “Everyone is equally sensitive, but death happens by chance once damage is high enough.”

Typical implications: Mortality curves are smooth and gradual. Well suited to pulse exposures and situations with delayed effects.

Analysis - Effects

GUTS - Survival

xAquaticRisk Analysis Outcome:

Excel-Table Output, 6 sheets:

- 3 species (Sp1, Sp2, Sp3)
- 2 GUTS models (IT, SD)

How to interpret:

For 10% of stream segments, in 25% of time,
survival rate is <89%
(Sp1=Asellus, SD model)

	A	B	C	D	E	F	G	H	I	J	K	L
1	perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100	
2	0.01	0.044	0.566	0.799	0.962	1.000	1.000	1.000	1.000	1.000	1.000	
3	0.05	0.070	0.601	0.817	0.967	1.000	1.000	1.000	1.000	1.000	1.000	
4	0.1	0.094	0.646	0.844	0.972	1.000	1.000	1.000	1.000	1.000	1.000	
5	0.25	0.168	0.745	0.891	0.984	1.000	1.000	1.000	1.000	1.000	1.000	
6	0.5	0.394	0.855	0.959	0.997	1.000	1.000	1.000	1.000	1.000	1.000	
7	0.75	0.759	0.951	0.988	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
8	0.9	0.905	0.989	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
9	0.95	0.919	0.993	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
10	0.99	0.934	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
11	1	0.938	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	
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Navigation bar: < > ... PECsw_percentiles_x_t | PECsed_percentiles_x_t | GUTSperc_SurvSdSp1 | GUTSperc_SurvSdSp2 | GU

Analysis - Effects

GUTS - Survival

spatial percentiles →

temporal percentiles ↓

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	0.639	0.899	0.960	0.993	1.000	1.000	1.000	1.000	1.000	1.000
0.05	0.660	0.916	0.965	0.994	1.000	1.000	1.000	1.000	1.000	1.000
0.1	0.690	0.923	0.973	0.996	1.000	1.000	1.000	1.000	1.000	1.000
0.25	0.754	0.955	0.985	0.999	1.000	1.000	1.000	1.000	1.000	1.000
0.5	0.883	0.985	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	0.967	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.9	0.985	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.95	0.991	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.99	0.994	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1	0.995	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

→ the short (1h) exposure of aquatic organisms in real-world stream systems seem to cause limited effects to invertebrates

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	0.922	0.985	0.994	0.998	1.000	1.000	1.000	1.000	1.000	1.000
0.05	0.928	0.988	0.995	0.998	1.000	1.000	1.000	1.000	1.000	1.000
0.1	0.938	0.989	0.996	0.998	1.000	1.000	1.000	1.000	1.000	1.000
0.25	0.953	0.993	0.997	0.999	1.000	1.000	1.000	1.000	1.000	1.000
0.5	0.982	0.997	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000
0.75	0.995	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.9	0.998	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.95	0.998	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.99	0.998	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1	0.998	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

SD – *Asellus aquaticus*

IT – *Asellus aquaticus*

Wetterau/Münster
arable, 1x7.5g/ha, April,
no mitigation

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.05	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.9	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.99	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

SD – *Cloeon dipterum*

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	0.998	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.05	0.998	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.1	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.25	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.9	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.99	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

IT – *Cloeon dipterum*

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	0.944	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.05	0.961	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.1	0.992	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.9	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.99	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

SD – *Gammarus pulex*

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	0.961	0.998	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.05	0.967	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.1	0.976	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.25	0.985	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.5	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.75	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.9	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.95	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
0.99	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

IT – *Gammarus pulex*

Analysis - Effects

GUTS - LP50 (Lethal Profile 50)

the **LP50** tells you how much you'd have to "scale up" an entire exposure pattern to reach 50% mortality.

LP50 = the **multiplication factor applied to a specific exposure profile** ($PEC_{sw}(t)$) that **results in 50% mortality** of the target organisms in a given period (eg, a year).

Illustration: you have a graph showing pesticide spikes in a stream segment over 30 days. "If these spikes were 5 times higher, would half the *Asellus* die?" If the model says yes, then your $LP50 = 5$.

In **regulatory context** (EFSA), the LP50 acts as a **Margin of Safety** (MoS).

If the LP50 is 100, the current environment is very safe (you'd need 100x more chemical to kill half the population).
If the LP50 is 1.2, you are uncomfortably close to a mass die-off.

Supplement / Analysis - Effects

GUTS - LP50

In Europe (EFSA), the **LP50 is the primary metric used when applying GUTS** as a higher-tier refinement in pesticide risk assessments.

To demonstrate that a pesticide is safe, the calculated LP50 must be equal to or greater than the Assessment Factor (AF) used in the lower tiers.

Organism Group	Assessment Type	Required LP50 (Trigger Value)
Aquatic Invertebrates	Acute (Survival)	LP50 $\geq$ 100
Fish	Acute (Survival)	LP50 $\geq$ 100
Chronic Refinement	Long-term Survival	LP50 $\geq$ 10

EFSA PPR Panel (2013): Guidance on tiered risk assessment for plant protection products for aquatic organisms in edge-of-field surface waters. This document established the "Tiered Approach," placing **GUTS under Tier 2C** (Refined Exposure).

EFSA PPR Panel (2018): Scientific Opinion on the state of the art of Toxicokinetic-Toxicodynamic (TKTD) effect models. The "bible" for GUTS in the EU. It officially declared GUTS models fit-for-purpose for regulatory use, provided they are properly validated for the specific substance and species.

The math behind this involves solving for a factor L in the survival function $S(t)$, such that at the end of the time period T : $S(T, L \times PEC_{sw}(t)) = 0.5$ where $PEC_{sw}(t)$ is the original exposure profile.

Analysis - Effects

GUTS – LP50

		spatial percentiles									
temporal percentiles	perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
	0.01	1.380	4.482	7.720	15.323	38.115	384.069	1000.000	1000.000	1000.000	1000.000
	0.05	1.468	4.745	8.621	16.446	40.201	457.311	1000.000	1000.000	1000.000	1000.000
	0.1	1.598	5.215	9.750	17.287	42.508	567.165	1000.000	1000.000	1000.000	1000.000
	0.25	2.012	7.179	12.536	20.674	56.307	1000.000	1000.000	1000.000	1000.000	1000.000
	0.5	3.780	12.395	18.453	29.741	91.213	1000.000	1000.000	1000.000	1000.000	1000.000
	0.75	7.818	20.572	25.497	43.568	201.365	1000.000	1000.000	1000.000	1000.000	1000.000
	0.9	14.414	27.359	36.085	67.134	554.307	1000.000	1000.000	1000.000	1000.000	1000.000
	0.95	17.220	30.282	38.777	84.554	777.153	1000.000	1000.000	1000.000	1000.000	1000.000
	0.99	17.926	33.055	40.377	92.406	953.488	1000.000	1000.000	1000.000	1000.000	1000.000
	1	18.055	33.717	40.807	94.180	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000

SD – Asellus aquaticus

→derive Tier-4 LP50 endpoints
eg, for the most sensitive species
and GUTS model

LP50_px10_pt25 = 12.5

Interpretation: for 25% of years and 10% of stream network, LP50 is <12.5. Thus, for all other times and stream segements LP50>12.5

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	10.620	27.686	46.513	103.507	259.791	1000.000	1000.000	1000.000	1000.000	1000.000
0.05	11.234	30.847	51.392	109.683	283.420	1000.000	1000.000	1000.000	1000.000	1000.000
0.1	11.595	32.873	57.601	116.575	296.413	1000.000	1000.000	1000.000	1000.000	1000.000
0.25	16.357	49.018	77.195	149.989	386.352	1000.000	1000.000	1000.000	1000.000	1000.000
0.5	26.455	78.457	127.952	220.882	602.706	1000.000	1000.000	1000.000	1000.000	1000.000
0.75	53.419	147.243	179.987	304.971	960.019	1000.000	1000.000	1000.000	1000.000	1000.000
0.9	77.352	193.098	261.031	435.234	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.95	95.293	214.870	297.492	567.040	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.99	103.242	238.297	311.022	598.962	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
1	106.431	241.122	314.237	602.324	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000

SD – Cloeon dipterum

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	1.485	3.944	7.077	14.316	34.820	438.145	1000.000	1000.000	1000.000	1000.000
0.05	1.546	4.275	7.541	15.126	37.997	564.003	1000.000	1000.000	1000.000	1000.000
0.1	1.722	4.616	8.165	15.775	41.292	746.622	1000.000	1000.000	1000.000	1000.000
0.25	1.992	7.132	11.293	20.789	54.485	916.753	1000.000	1000.000	1000.000	1000.000
0.5	3.784	11.322	16.940	29.307	84.637	1000.000	1000.000	1000.000	1000.000	1000.000
0.75	5.937	18.871	24.439	40.210	158.663	1000.000	1000.000	1000.000	1000.000	1000.000
0.9	10.871	26.948	34.206	65.993	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.95	13.877	29.056	37.174	79.240	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.99	14.680	30.671	39.258	86.774	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
1	14.893	31.191	39.947	89.197	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000

SD – Gammarus pulex

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	4.795	15.202	26.479	50.731	128.283	1000.000	1000.000	1000.000	1000.000	1000.000
0.05	5.119	16.709	28.956	54.344	135.158	1000.000	1000.000	1000.000	1000.000	1000.000
0.1	5.596	17.690	32.383	58.545	144.674	1000.000	1000.000	1000.000	1000.000	1000.000
0.25	6.837	24.870	42.293	68.370	191.085	1000.000	1000.000	1000.000	1000.000	1000.000
0.5	12.659	42.812	61.229	101.029	302.247	1000.000	1000.000	1000.000	1000.000	1000.000
0.75	30.435	68.573	86.109	145.715	669.171	1000.000	1000.000	1000.000	1000.000	1000.000
0.9	48.088	88.407	120.565	216.539	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.95	56.928	98.515	128.266	278.658	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.99	58.929	109.230	133.030	303.587	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
1	59.344	112.432	134.208	311.674	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000

IT – Asellus aquaticus

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	40.369	91.021	149.815	353.525	849.947	1000.000	1000.000	1000.000	1000.000	1000.000
0.05	41.202	97.851	164.403	371.618	879.659	1000.000	1000.000	1000.000	1000.000	1000.000
0.1	42.760	102.454	181.197	400.067	967.944	1000.000	1000.000	1000.000	1000.000	1000.000
0.25	55.619	153.255	242.327	522.496	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.5	101.229	243.415	416.534	744.003	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.75	189.053	490.177	623.177	974.722	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.9	265.433	654.130	866.884	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.95	345.806	718.765	916.501	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.99	362.070	795.234	978.670	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
1	364.988	810.962	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000

IT – Cloeon dipterum

perc_x: rig	Px1	Px5	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
0.01	2.859	8.353	14.220	27.375	73.017	744.174	1000.000	1000.000	1000.000	1000.000
0.05	3.099	9.121	15.613	29.469	76.184	841.818	1000.000	1000.000	1000.000	1000.000
0.1	3.349	9.947	17.608	31.849	80.331	961.015	1000.000	1000.000	1000.000	1000.000
0.25	4.030	13.999	22.874	39.239	102.740	1000.000	1000.000	1000.000	1000.000	1000.000
0.5	7.197	22.644	33.909	56.432	166.456	1000.000	1000.000	1000.000	1000.000	1000.000
0.75	12.006	38.172	47.040	78.035	334.843	1000.000	1000.000	1000.000	1000.000	1000.000
0.9	24.180	48.199	65.770	127.004	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.95	30.233	53.904	70.357	154.050	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
0.99	31.466	59.690	72.220	163.851	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000
1	31.774	61.091	72.515	164.624	1000.000	1000.000	1000.000	1000.000	1000.000	1000.000

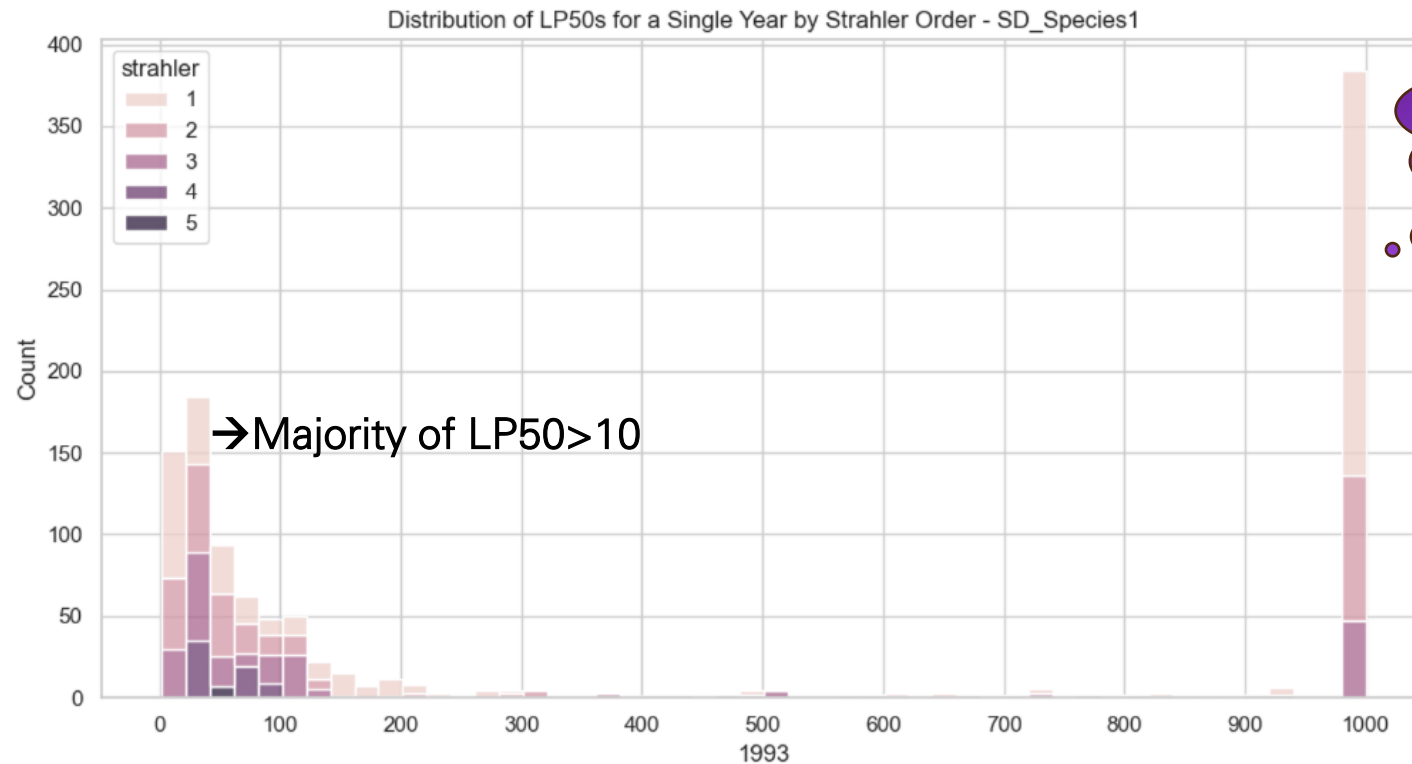
IT – Gammarus pulex

Analysis - Effects

GUTS – LP50

LP50 distribution over stream segments

for the most sensitive species (Asellus aq.) and the most sensitive GUTS model (SD)
(by stream Strahler order, single year, threshold cut by 1000)

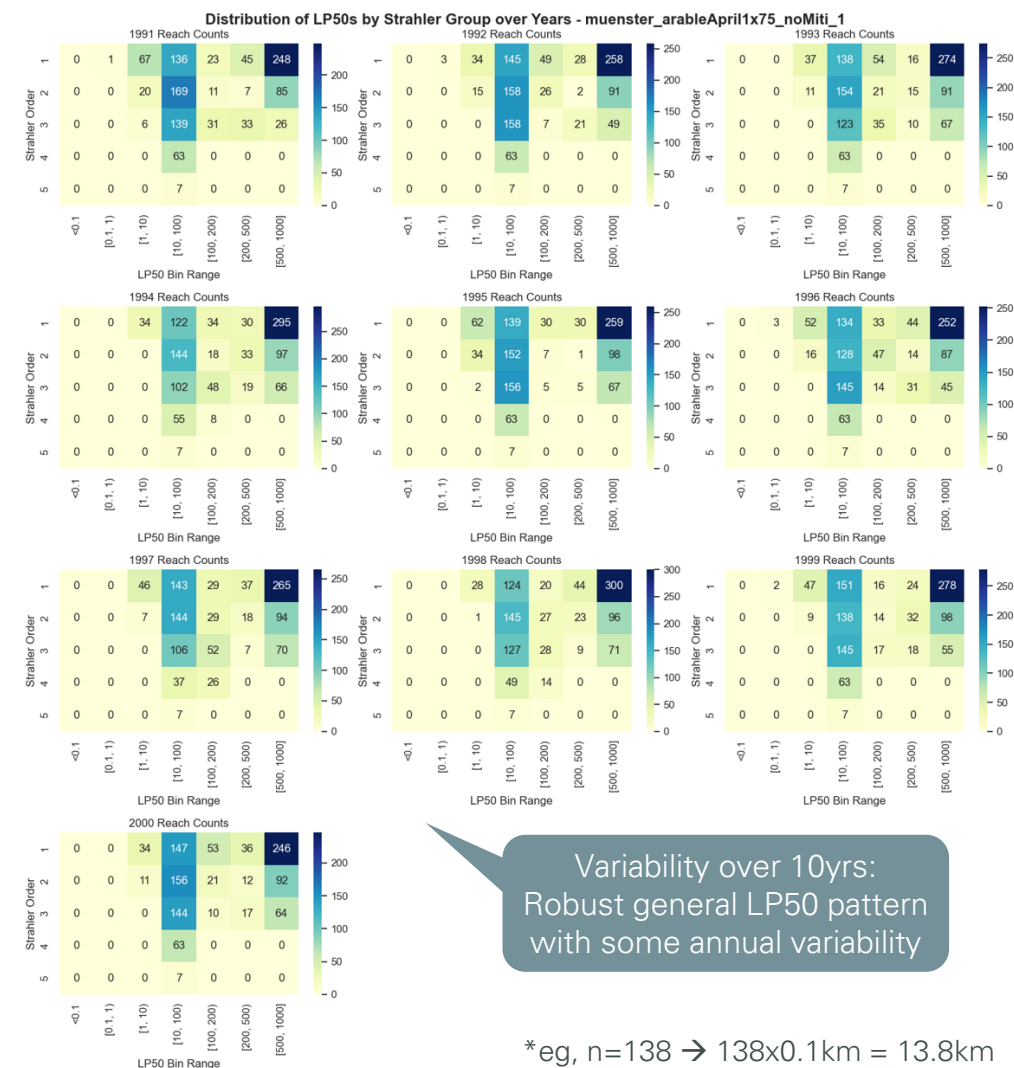
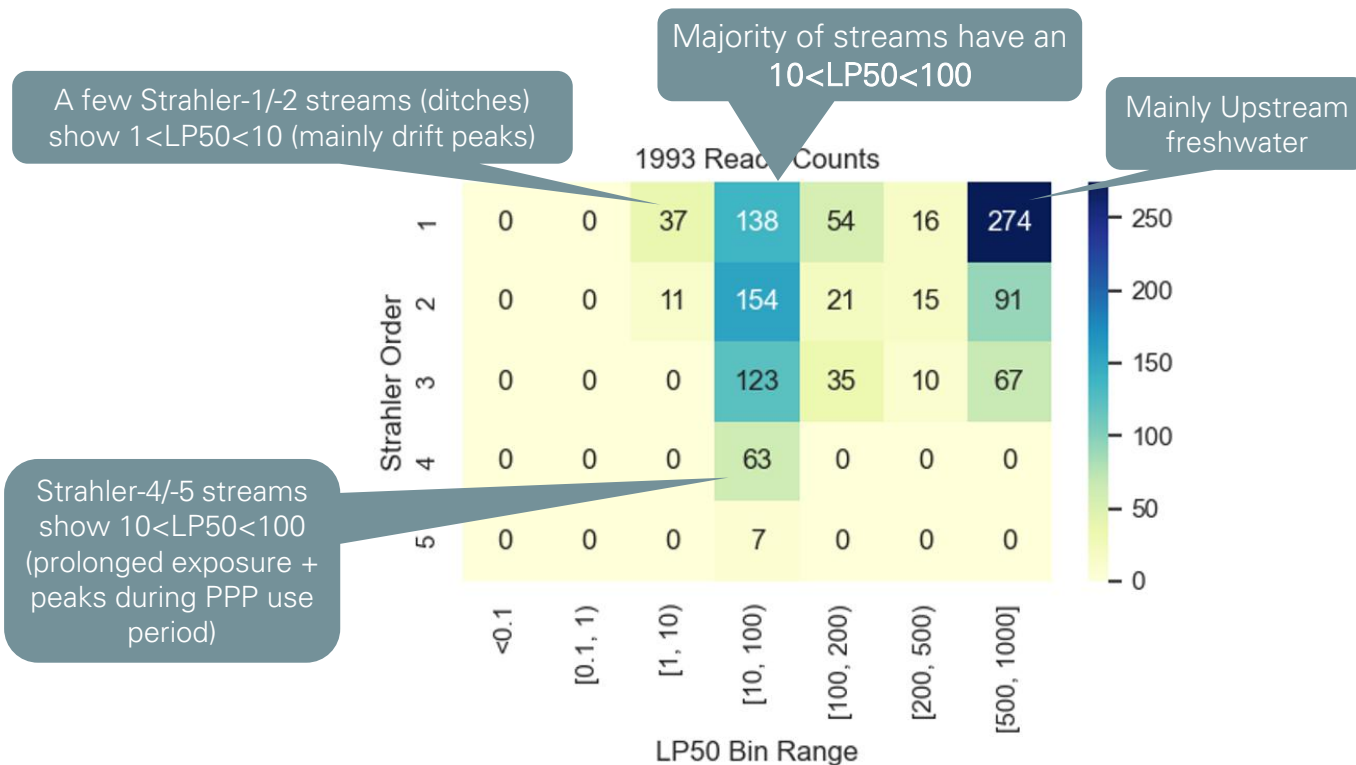


Which stream segments show a LP50 > 1000?

Analysis - Effects GUTS – LP50

LP50 distribution over stream segments

reach counts, by Strahler order, by year, by LP50 bin group
Absolute count numbers* allow for an **ecological impression of effect extents** (color pattern is similar for relative numbers)



Wetterau/Münster arable, 1x7.5g/ha, April, no mitigation
most sensitive species (*Asellus aq.*), most sensitive GUTS model (SD)

*eg, $n=138 \rightarrow 138 \times 0.1\text{km} = 13.8\text{km}$

Analysis - Effects GUTS – LP50

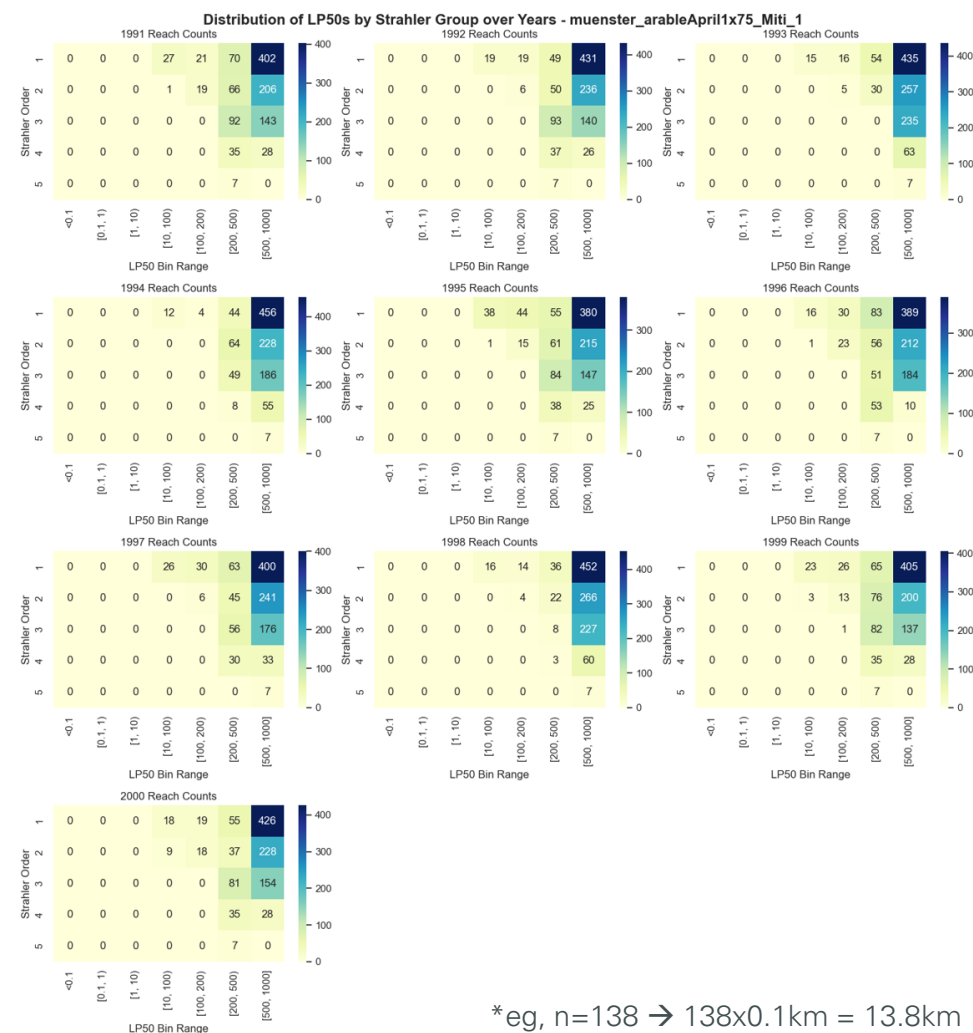
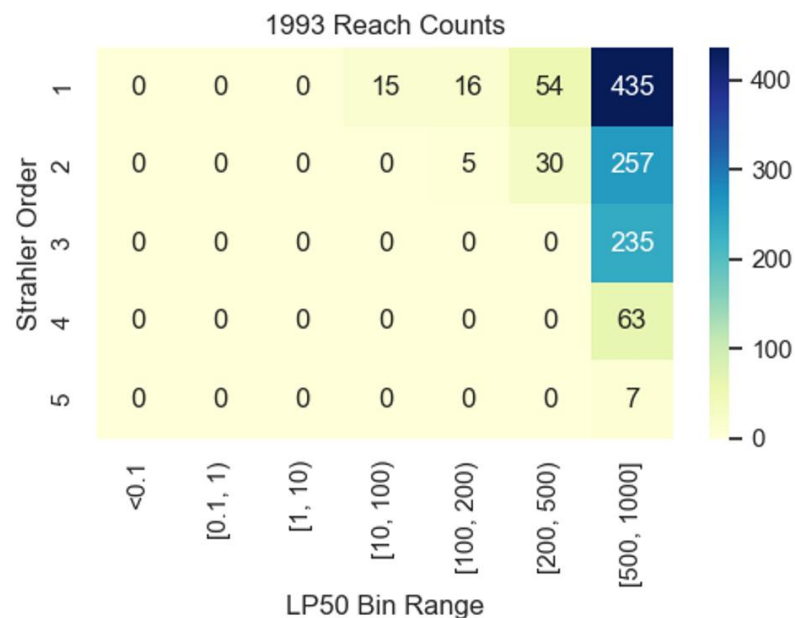
LP50 distribution over stream segments

reach counts, by Strahler order, by year, by LP50 bin group

Absolute count numbers* allow for an **ecological impression of effect extents** (color pattern is similar for relative numbers)

Apply mitigation:

The majority of streams have an
LP50 > 200



Wetterau/Münster

arable, 1x7.5g/ha, April, **mitigation: 20m buffer, 50% drift reduction**

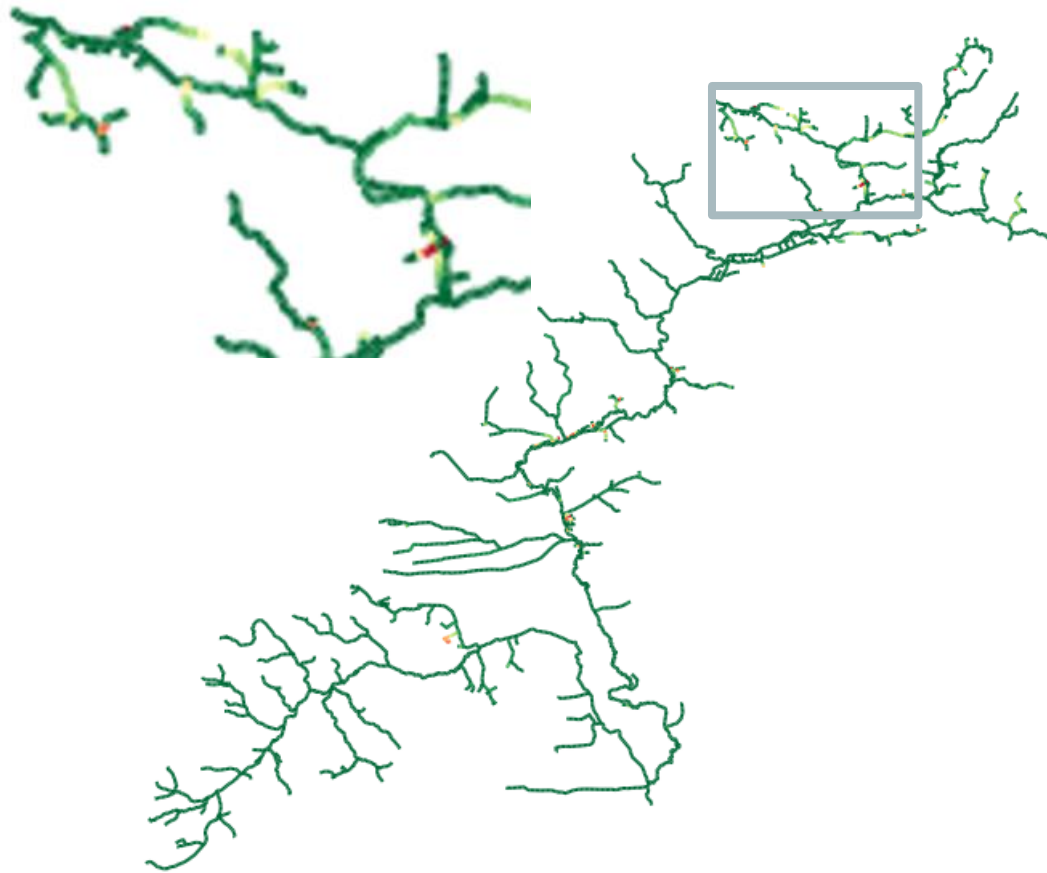
most sensitive species (*Asellus aq.*), most sensitive GUTS model (SD)

*eg, n=138 → 138x0.1km = 13.8km

Analysis – Effects Mapping

no mitigation

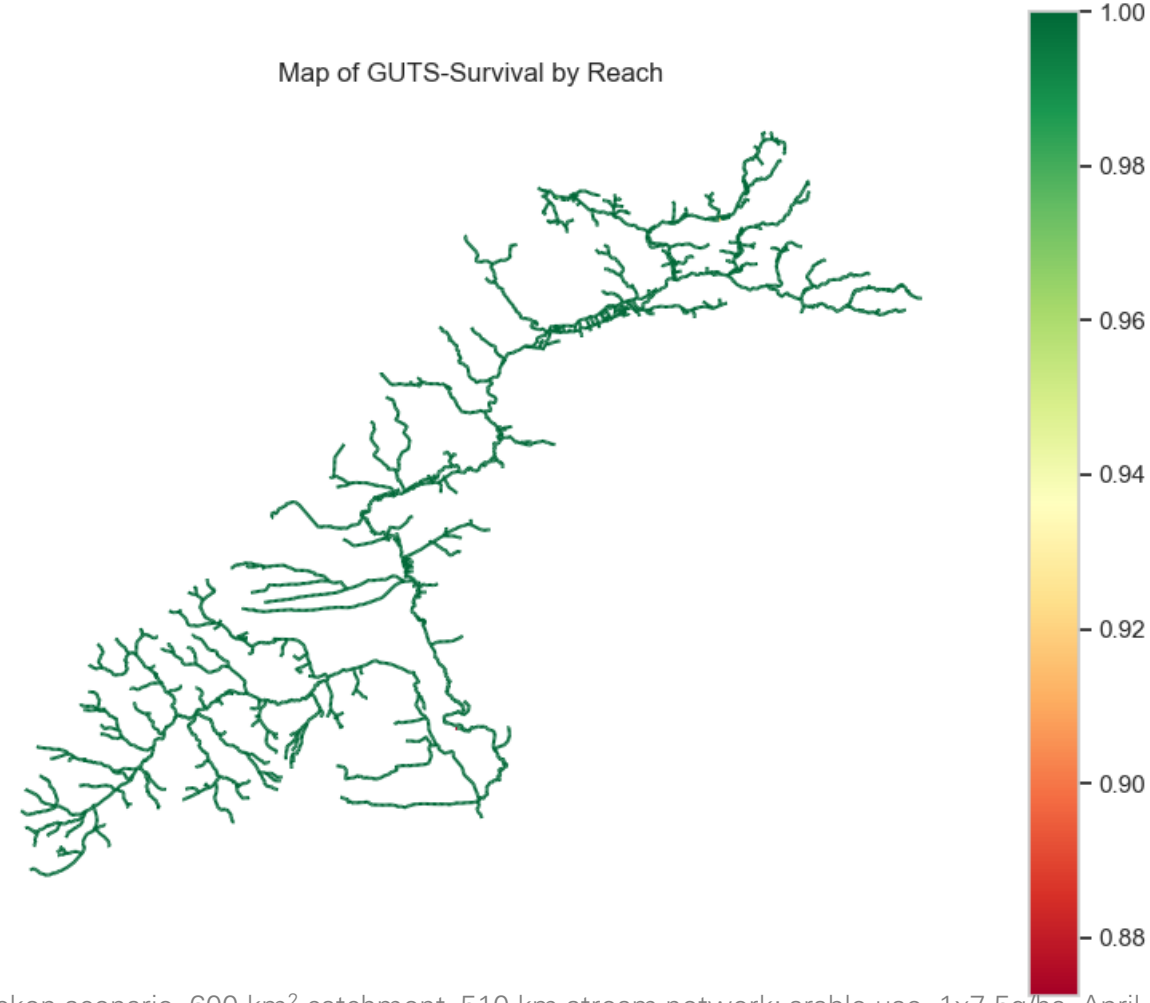
Map of GUTS-Survival by Reach



SD – *Asellus aquaticus*

mitigation: 20m buffer, 50% drift reduction

Map of GUTS-Survival by Reach



Bruchenbrücken scenario, 600 km² catchment, 510 km stream network; arable use, 1x7.5g/ha, April

xAquaticRisk UseCase

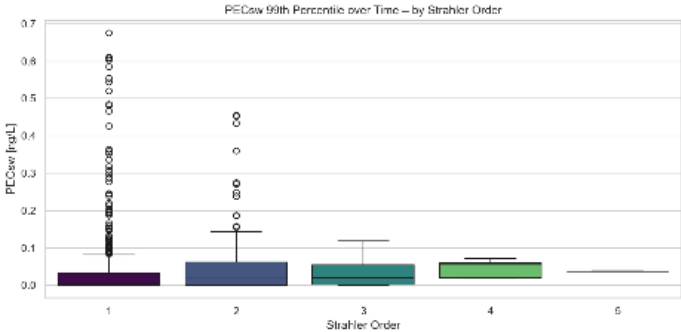
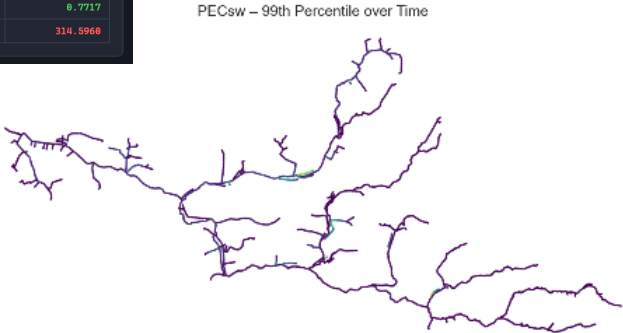
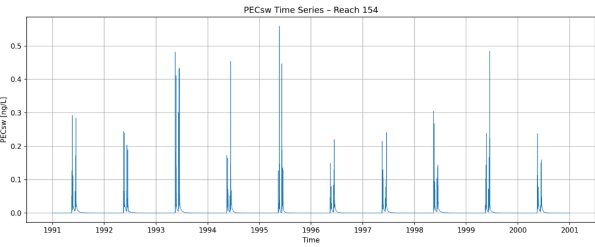
Model Outcome Summary

Build a higher-tier
Risk A Story

Exposure&Effects

PECsw Percentile Table [ng/L] — rows: temporal percentiles, columns: spatial percentiles

pt / px —	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
Pt10	0	0	0	0	0	0	0	1.00e-4
Pt25	0	0	0	0	0	0	0	4.00e-4
Pt50	0	0	0	0	1.00e-4	1.00e-4	1.00e-4	0.0027
Pt75	0	0	0	2.00e-4	5.00e-4	6.00e-4	9.00e-4	0.0156
Pt90	0	0	1.00e-4	0.0014	0.0039	0.0048	0.0077	0.0851
Pt95	0	0	5.00e-4	0.0042	0.0135	0.0151	0.0209	0.1969
Pt99	0	0	0.0050	0.0422	0.0796	0.1153	0.1747	0.7717
Pt100	0	0	1.0285	4.8364	18.4861	39.5371	133.5139	314.5968



Next: Population Effects

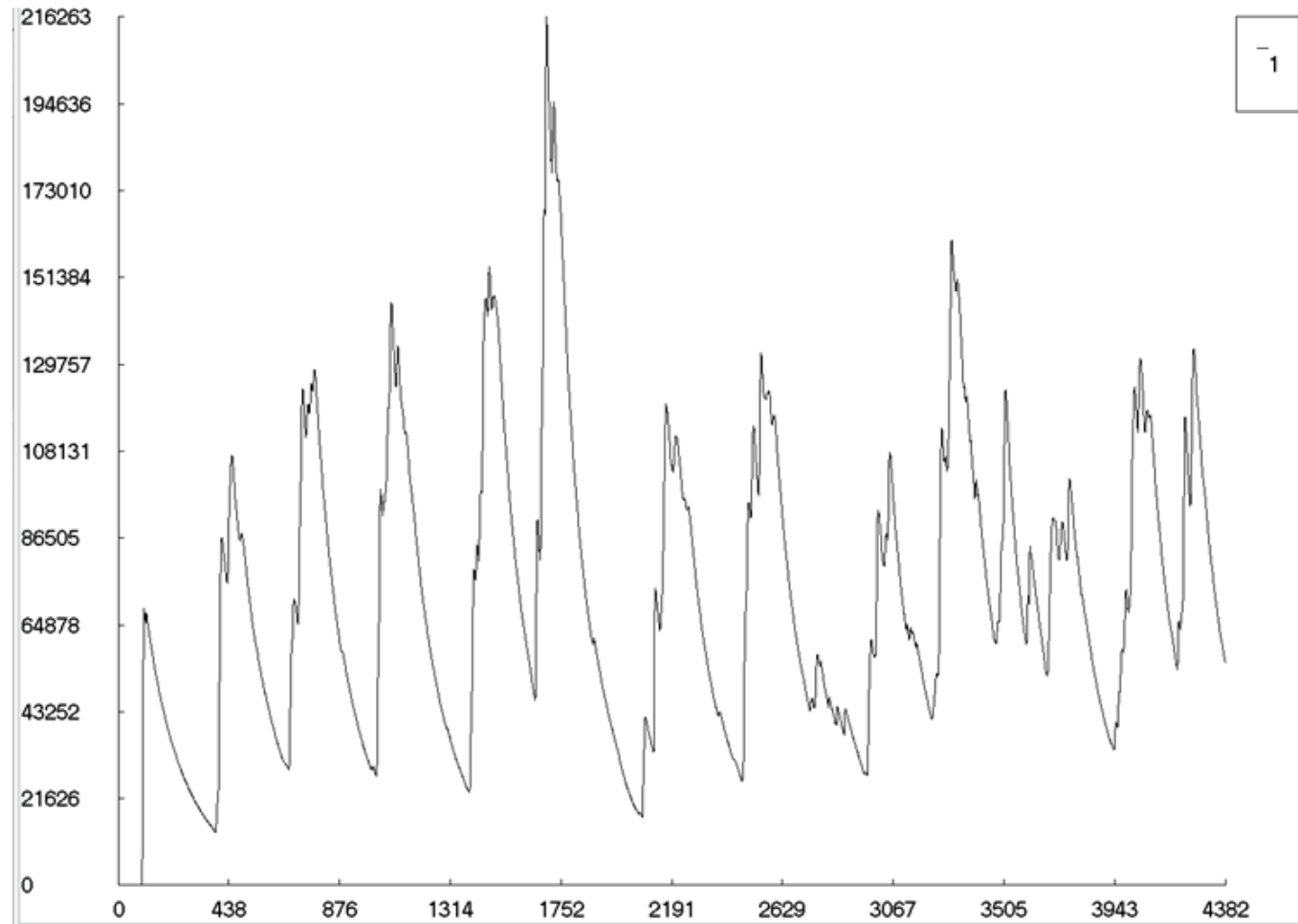
LPOP Model → xAquaticRisk Component

Example **Asellus aquaticus**

Evaluation of **effects on population level**
(Ecological Threshold Option, ETO)

Evaluation of **recovery potential**
(Ecological Recovery Option, ERO)

LPOP Component already integrated and used in xAquaticRisk. In-depth review and scientific publication ahead.





Results & endpoints based on biological perspective

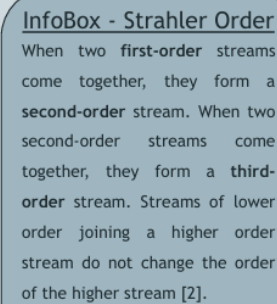
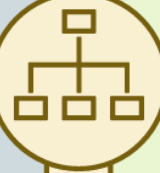
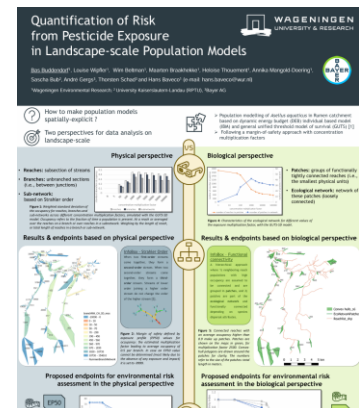
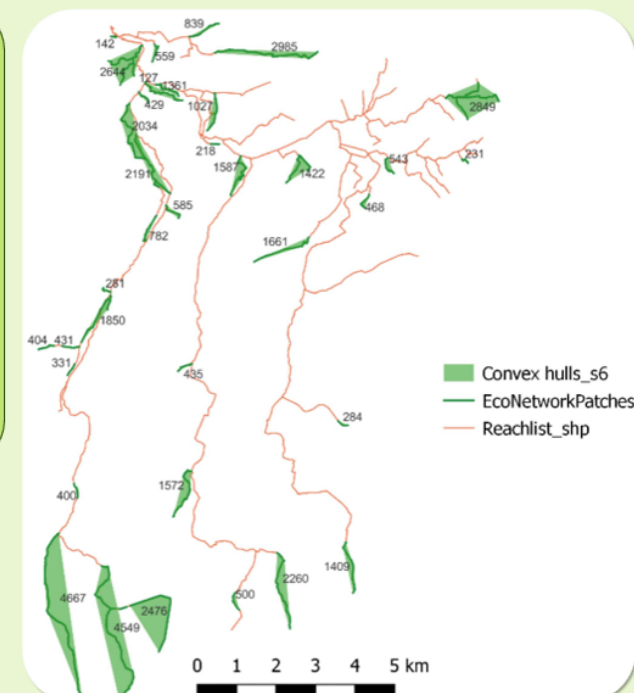


Figure 2: Margin of safety defined by exposure profile (EP50) values for occupancy: the estimated multiplication factor leading to average occupancy of 0.5 per branch. In case an EP50 value cannot be determined (most likely due to the absence of any exposure and impact) it is set to -9999.



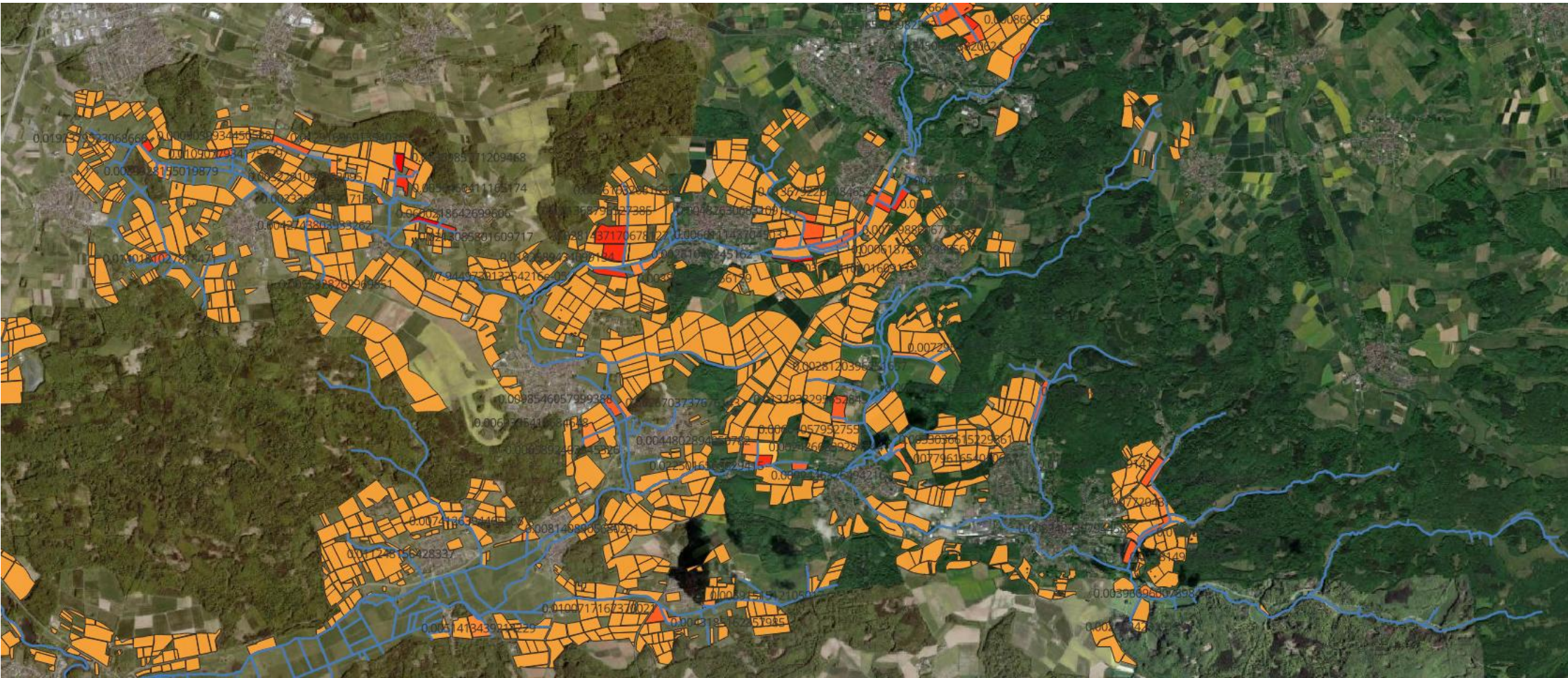
A hierarchical approach where 1) neighboring reach populations with high occupancy are assumed to be connected and are grouped in **patches**, and 2) patches are part of the **ecological network** and functionally connected depending on species dispersal attributes.

Figure 5: Connected reaches with an average occupancy higher than 0.9 make up patches. Patches are shown on the maps in green, for multiplication factor 2500. Convex-hull polygons are drawn around the patches for clarity. The numbers refer to the size of the patches: total length in meters.



Next: Analysis – Risk Management Mapping

eg, AI: 'show me the most drift contributing fields...'



Next

Community

Users and developers. Connect to research, development and risk assessment teams sharing the demand, thinking and working along the same lines. Trainings. Workshops. Support.

Readyness and Access

Documentation of xLandscape, Components and Scenarios. Testing.
Simple download and getting started. Simple access to case studies, eg via web.
Scenario development.

Adapt AI

AI support and integration in all aspects, eg, building Components, Landscape Models and Scenarios.
Model parameterisation (easy ai-based and formal access to pesticide data), execution and analysis, report writing, etc.

Development

of Components, Models and Scenarios. Maintenance.
xLandscape Version-4: clean architecture (microkernel and component-based software), improved autonomy of components, Digital-twin readiness, coupling of monitoring and modelling for systems-approaches.





xAquaticRisk Modelling take home

- (1) **xAquaticRisk is a ready-to-use landscape (catchment) model** to simulate pesticide use, **exposure** of aquatic habitats and **effects on individual and population level**. It is a **modular** composition of individual models. It operates **spatiotemporally explicit**.
- (2) In Europe, risk assessment is designed as a **tiered approach** and defines **Specific Protection Goals**. **xAquaticRisk can deliver exposure and effect endpoints** that serve risk assessments. Endpoints are embedded into a (real-world) **context of exposure and effects**, as supportive information for higher-tier, **weight-of-evidence** risk assessment.
- (3) **xAquaticRisk is applied as any other model**, providing parameterization and scenarios. Some default **analysis tools** of exposure and effects are pre-built. Any data science approaches can be used in the analysis (including AI).
- (4) xAquaticRisk is available on **Github** (<https://github.com/xlandscape/xAquaticRisk>) and can be download for **copy-and-run**. Its **development is continued** to broaden its applicability to a range of problems and research questions.

Our Values

Thanks!

Openness & Collaboration

We believe in **open source** principles, making our models and components accessible for everyone. This **fosters innovation**, enables **participation**, and ensures that solutions are tailored to real-world challenges rather than built in isolation.

FAIR Principles

We follow the **FAIR guidelines**—Findable, Accessible, Interoperable, and Reusable. By designing modular frameworks and **sharing transparently**, we empower researchers and stakeholders to build on existing work and accelerate progress.

Invitation to Participate

xLandscape development come with a **mindset**. We **invite everyone** to contribute ideas, share expertise and help shaping the future of landscape modelling for sustainable development.

thorsten.schad@landwerk-ev.de



Supplement / Concept Modularity enables...



Diverse Topics and Competences
on landscape level

Collaboration

Focus on key expertise
Build on existing work



Harmonisation

use similar solutions to similar problems

Transparency

Flexibility – adapt model complexity

Consistent Integration

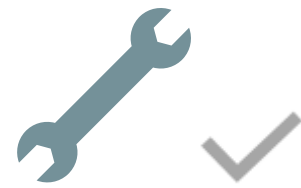
Stepwise
Validation

use established models
moduls can become certificated APIs*



Appendix





Supplementary Exercises

- 1) Run experiments without and with different mitigation measures. Which mitigation is most effective in view of aquatic risk? Which mitigation is most effective in holistic view, taking cost for crop production into account?
- 2) Develop visualisation and spatial analysis using the GIS (eg, QGIS)
- 3) Imposing a buffer mitigation of 20m, which fraction of arable land cannot be treated, which fraction of individual fields cannot be treated, which fields are most affected? How many fields (which fraction of fields) are unaffected?
- 4) Write an analysis that calculates the total amount of the ModelPyrethroid in the total system (water phase and sediment)
- 5) Tell us your wishes for the xAquaticRisk Model. Eg, for improvements on the ControlPanel, Analysis, Components, Exposure routes, Effect models, your applications, Trainings, tutorials, workshops, online training...
send to: thorsten.schad@landwerk-ev.de

Questions, Discussion Points

1. Why move to higher-tier/Tier4? FOCUS gives ~2 ng/L while the RAC is ~1 ng/L (unacceptable at lower tier). What exactly would you need to see (data or model evidence) to justify moving to a Tier-4 landscape approach – and what would count as “more realistic” in this case?
2. SPGs as the steering wheel. Which SPG “dimension” is most decisive for choosing endpoints in this training – entity, attribute, magnitude, temporal scale, spatial scale, or certainty – and why? (Pick one and defend it.)
3. ETO vs ERO (threshold vs recovery). Which extra efforts will the ERO bring to higher-tier landscape modelling approach for an insecticide.
For pyrethroids and aquatic invertebrates, when would you argue for an Ecological Threshold Option (ETO) versus an Ecological Recovery Option (ERO)? What kind of evidence would you consider “strong enough” to support ERO?
4. What changes from FOCUS → Tier-4 (and what doesn't)? What is fundamentally different when going from a Step 3/4-style exposure–RAC comparison to a Tier-4 landscape coupling of exposure and effects – and what conservative assumptions may still remain even at Tier-4?
5. The “ensemble” idea (protecting the system, not every ditch). The slides argue for protecting an ensemble of edge-of-field waters rather than each single segment. How would you explain (to a regulator or stakeholder) why that is still protective – and where you would draw the line for unacceptable hotspots?
6. Endpoints & percentiles: what do they mean in practice? When you see “spatial percentiles” and “temporal percentiles” of $PEC_{sw}(x,t)$, what real-world protection statement are you making? Which percentile combination feels defensible for decision-making – and why?
7. Exposure framing choice: “Exposed-only streams”. What changes (conceptually and numerically) when you switch to an “exposed-only streams” analysis? In which regulatory contexts could this be justified, and where could it be misleading?
8. Mitigation: from simulation result → practical rule. The mitigation example (buffer + drift reduction) shows large reductions. How would you translate that into a simple, auditable risk-management rule (e.g., targeted hotspots vs blanket mitigation), and what extra information would you need?

There are **no single correct answers**^{**}; the goal is to surface assumptions and trade-offs

Higher-Tier Aquatic Risk Assessment

EFSA has been working on:
 updated **specific protection goals**,
 revised **ecological recovery concepts**,
 new **TK/TD integration**,
 and harmonisation with the **2021–2023 exposure guidance updates**.

Apply newer scientific methods (e.g., TK/TD, population models) only as refinements, not replacements.

EFSA PPR Panel (EFSA Panel on Plant Protection Products and their Residues), 2013. Guidance on tiered risk assessment for plant protection products for aquatic organisms in edge-of-field surface waters. EFSA Journal 2013;11(7):3290, 268 pp. doi:10.2903/j.efsa.2013.3290.

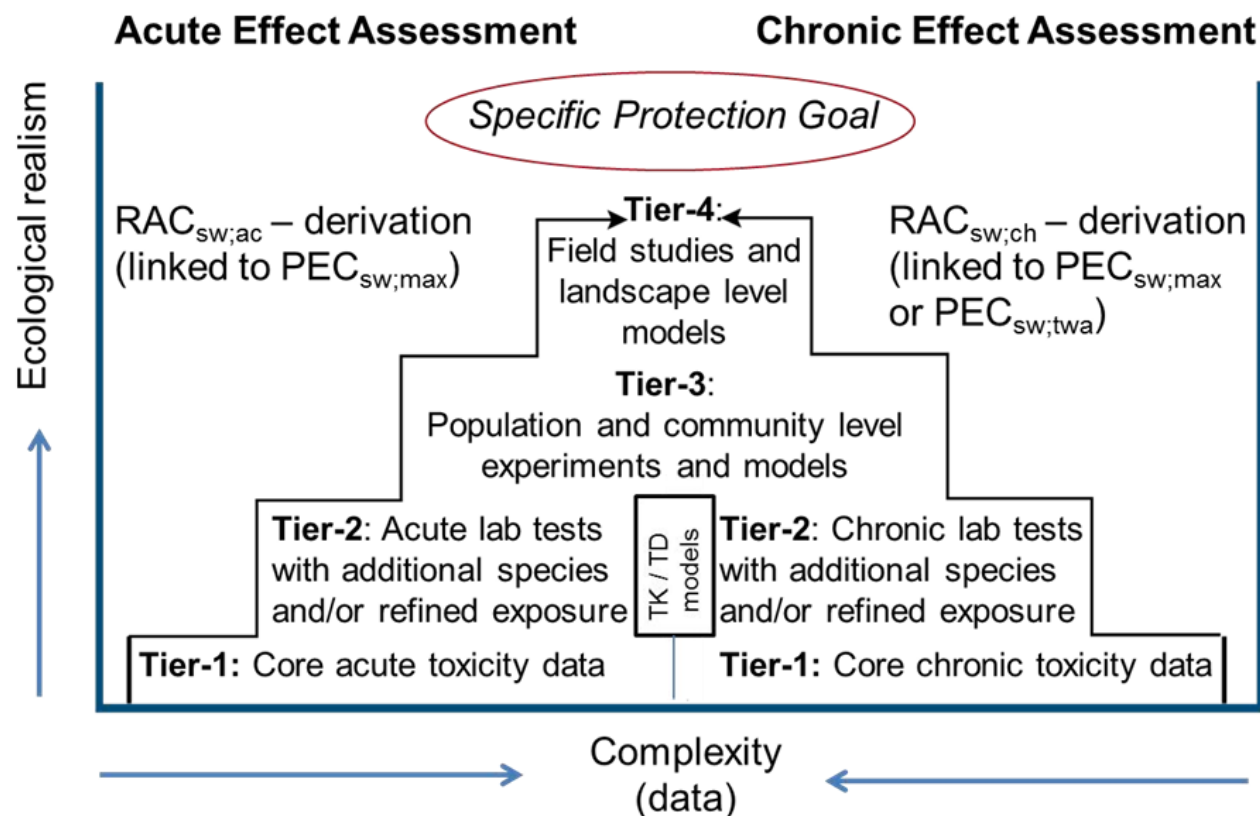
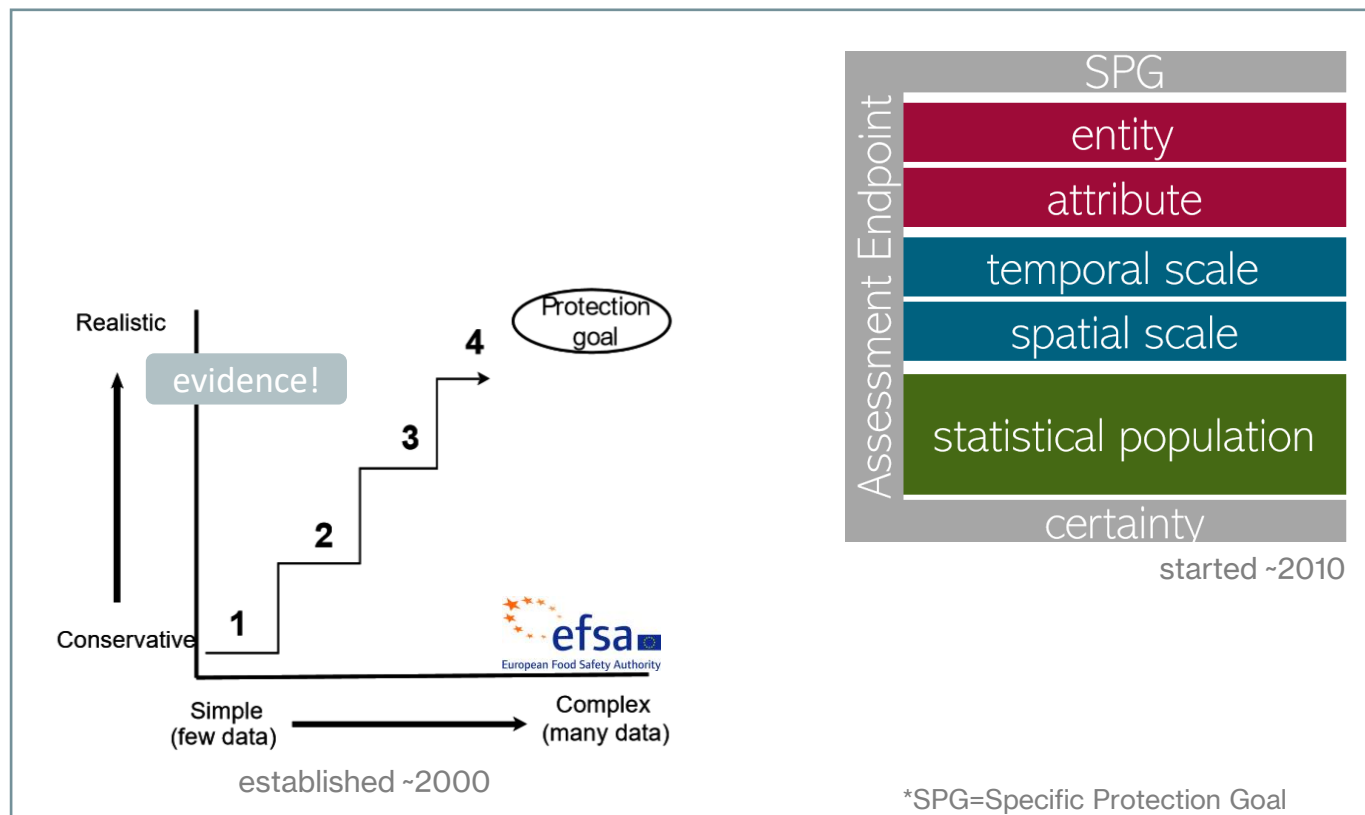


Figure 1: Schematic presentation of the tiered approach within the acute (left part) and chronic (right part) effect assessment for plant protection products (PPPs). For each PPP, both the acute and chronic effects/risks have to be assessed. The tier 1 and tier 2 effects assessments are based on single species laboratory toxicity tests, but to better address risks of time-variable exposures the tier 2 assessment may be complemented with toxicokinetic/toxicodynamic (TK/TD) models. Tier 3 (population- and community-level experiments and models) and tier 4 (field studies and landscape-level models) may concern a combination of experimental data and modelling to assess population- and/or community-level responses (e.g. recovery, indirect effects) at relevant spatio-temporal scales. All models included in such a tiered approach need to be properly tested and fulfil required quality criteria.

Higher-Tier aquatic RA

Here: Primary regulatory target: Threshold Option (recovery option addressed as supplementary information, eg, hot-spot pattern, connectivity)



Key concepts driving the xLandscape design:

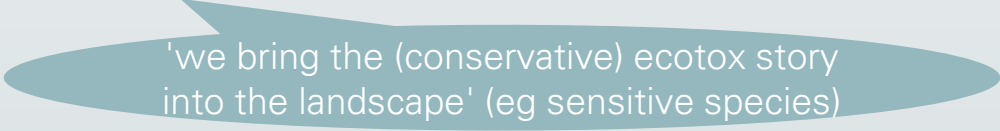
- **Tiered approach** → flexibility, **adaptivity**, *'get only as complex as necessary'*
- **Specific Protection Goals** → operate in explicit **dimensions** and **scales** (spatial, temporal), explicit **structure** of analysis and communication

Supplement

Tier4 Design / Conservative Elements

Conservative elements: Although more realistic, even a **landscape-scale RA** is a model system and still contains a number of conservative assumptions, eg,

- **100% product use** (DLT) in all crops given by the potential label, dh essentially on **100% of fields**
- **No spray-drift filtering** by riparian vegetation (still 'filter-paper' spray-drift deposition scenario).
- **No drying-out** of small ditches. This causes (artificially?) high peak PEC_{sw}, eg, at 1cm minimum water level
- **100% of sensitive organisms exposed** (no spatial heterogeneity in stream system). **100% occurrence** of sensitive species
- ...



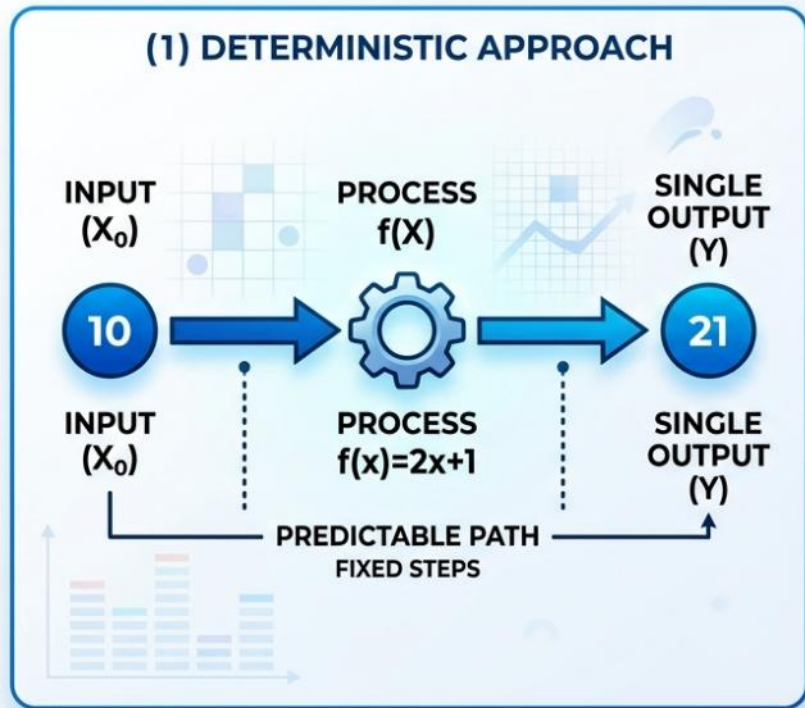
'we bring the (conservative) ecotox story into the landscape' (eg sensitive species)

Supplement

Deterministic vs Probabilistic

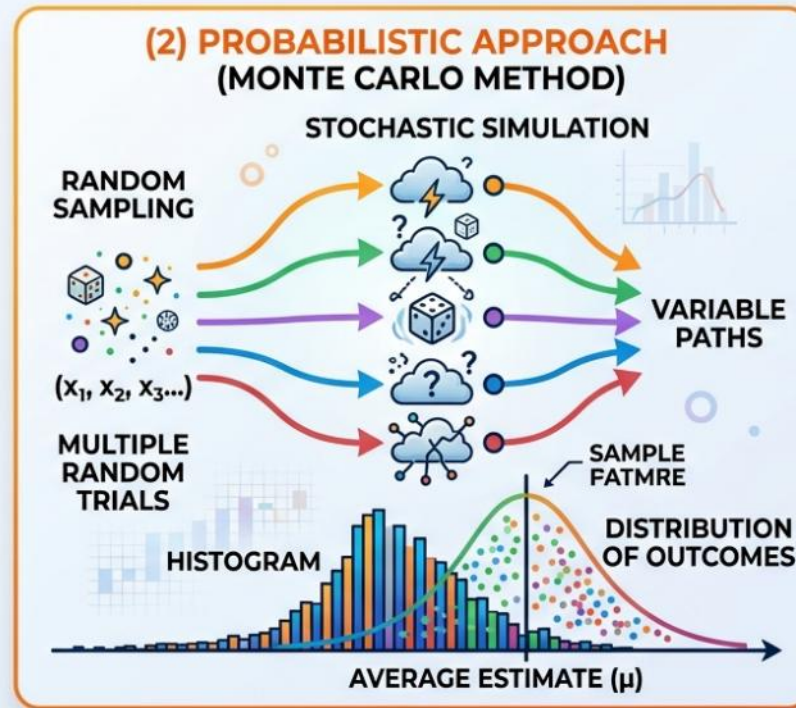
ILLUSTRATING THE MONTE CARLO METHOD: DETERMINISTIC VS. PROBABILISTIC APPROACHES

(1) DETERMINISTIC APPROACH



Fixed Inputs -> Fixed Formula -> Single, Fixed Outcome.
No Uncertainty.

(2) PROBABILISTIC APPROACH (MONTE CARLO METHOD)



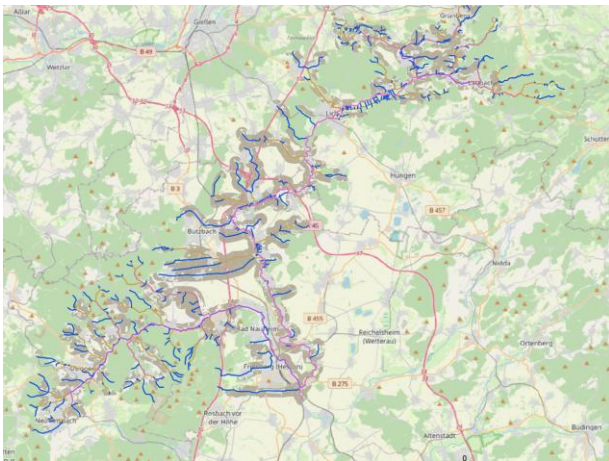
Random Inputs -> Simulation trials -> Distribution of Outcomes. Estimated mean value.

Study Representativeness

- Scenario site selection / region-of-interest (ROI) representativeness is key for working within the regulatory framework
- Study is representative analysis is guided by its purpose (eg, general scientific study, conservation, landscape design, regulatory risk assessment, risk management)
- Demonstrate that the study has covered the range of real-world conditions, esp. 'upper-end' risk conditions in regulatory risk assessment
- Often operates with risk indicators (eg, metamodels, expert judgement, literature) on larger geographic regions
- Approaches, eg
 - sampling of sites; grid; indicator-based; manual selection of conservative grids + random grids (open, extendable)
 - full coverage of all crop cultivation for (spatially) limited crops (eg, hops in Germany, orchards)
 - distinct site selection guided by further reasons, eg, known region of interest (eg, water extraction, long-term scientific study site; pilot region for testing new authorisation principles)

Available Scenarios (April 2026)

Catchment	Sub-Catchment	Origin	Stream Length [km]	Area [km ²]	Focal Crop(s)
Rummen	Rummen	xAquatic dev.	170	190	apple
	Oudebeek	xAquatic dev.	15	10	apple
Wetterau	Muenster	Master thesis	110	90	arable
	Muschenheim	Master thesis	195	180	arable
	Bruchenbrücken	Bayer (knoell)	510	600	arable
Funne	Funne	BASF	56	70	arable
Grote Kemmelbeek	Grote Kemmelbeek	Bayer (monitoring, risk mgmt project)	16	11	arable



Supplement

Focus Analysis to 'Exposed-only' Stream Segments

'Leaving the catchment behind'

ie, more classical conservative RA focusing on stream segments and times of exposure (PEC_{sw}>0).

Excludes especially upstream reaches and initial simulation period until first PPP use.

		spatial percentiles				PEC _{sw} (x(t)) [ng/L]			
temporal percentiles	perc_x: rig	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
	0.1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001
	0.25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0004
	0.5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0016
	0.75	0.0000	0.0000	0.0000	0.0001	0.0003	0.0003	0.0014	0.0104
	0.9	0.0000	0.0000	0.0003	0.0009	0.0017	0.0024	0.0088	0.0475
	0.95	0.0000	0.0001	0.0010	0.0034	0.0062	0.0068	0.0303	0.1196
	0.99	0.0004	0.0014	0.0099	0.0272	0.0487	0.0577	0.2765	0.4531
	1	0.1699	0.4640	1.2804	5.4877	16.9820	31.7659	94.9339	203.1640

		spatial percentiles				PEC _{sw} (x(t)) [ng/L]			
temporal percentiles	perc_x: rig	Px10	Px25	Px50	Px75	Px90	Px95	Px99	Px100
	0.1	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001
	0.25	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0003
	0.5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0015
	0.75	0.0000	0.0000	0.0000	0.0001	0.0002	0.0003	0.0007	0.0095
	0.9	0.0000	0.0000	0.0001	0.0006	0.0016	0.0018	0.0067	0.0467
	0.95	0.0000	0.0000	0.0004	0.0023	0.0057	0.0062	0.0236	0.1169
	0.99	0.0000	0.0001	0.0043	0.0208	0.0424	0.0511	0.2520	0.4420
	1	0.0000	0.1059	0.7526	3.6506	13.8394	25.9181	75.9850	203.1640

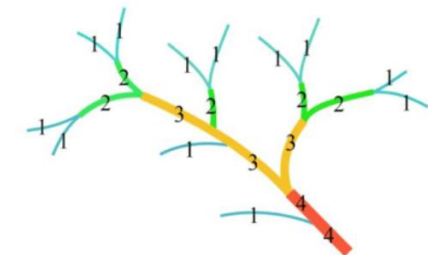
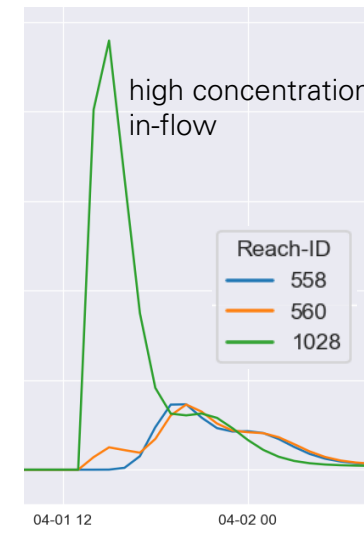
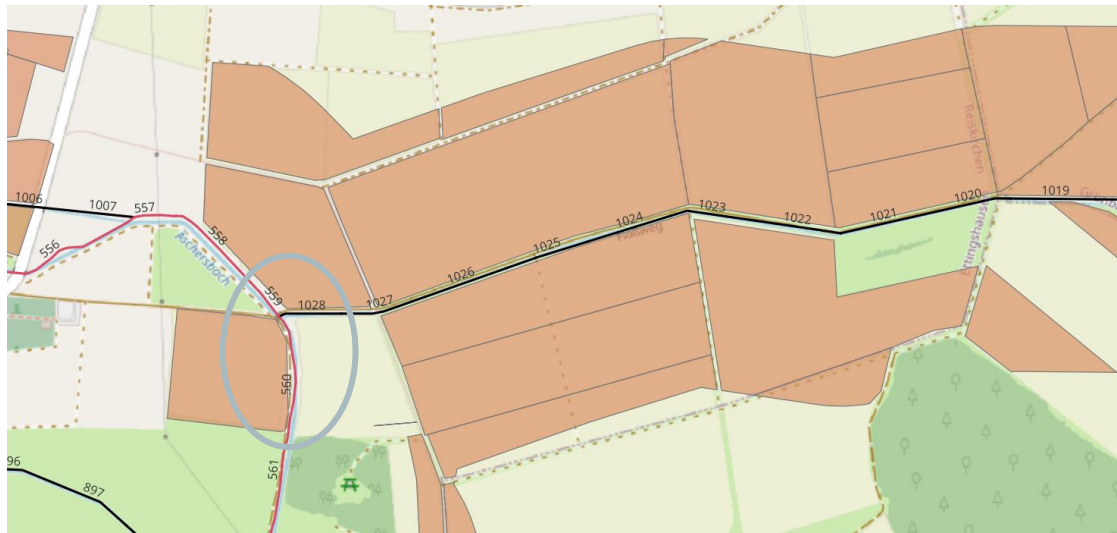
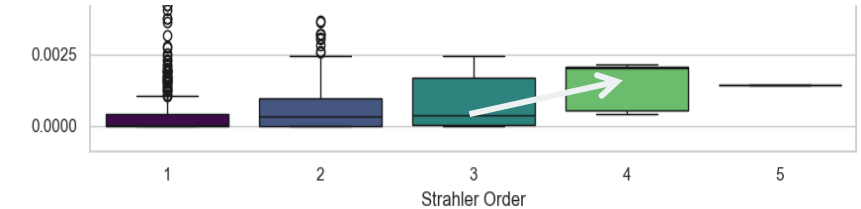
→ slightly higher
PEC_{sw} percentiles

Outcome – Tier-4 / PEC_{sw} by Strahler

(Evaluation Reliability Detail)

What seems counter-intuitive in the first place is correct: higher-order stream segments can have higher concentrations than such of lower order, esp. for rare temporal events (percentiles)

- if lower strahler order segments (eg, Strahler 1) of high concentrations flow into such of higher order (eg, Strahler 4) (fig 1)
- or if higher order segments (eg, Strahler 4) have direct exposure entries (fig 2)



Outcome – Tier-4 / PEC_{sw} by Strahler

(Evaluation Reliability Detail)

✓ what seems **counter-intuitive** in the first place is correct: higher-order stream segments can have higher concentrations than such of lower order, esp. for rare temporal events (percentiles)

- if lower strahler order segments (eg, Strahler 1) of high concentrations flow into such of higher order (eg, Strahler 4) (fig 1)
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